



**CE3100 Controller
for CHECKWEIGHER**

Modbus/TCP Communication

User's manual

First Edition

CB45228H0070

Yamato Scale Co., Ltd.

CE3100 Controller Modbus/TCP Communication User's manual

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Yamato Scale Co., Ltd.

PREFACE

This instruction manual provides operational procedures of the controller (model CE3100) for YAMATO auto checkweighers with regard to the Modbus/TCP Communication along with some safety instructions. Other operation manuals are provided for the auto checkweigher. All the personnel who carry out the operation and maintenance work must refer to the designated chapters of those manuals.

- Prior to installation, operation and maintenance work of the machine, carefully read the related chapters of those manuals.
- Keep the instruction manuals on hand so that they can be referred to at any time.
- YAMATO shall not be liable for any problems caused by the user's disregard of the procedures or instructions specified in this manual or other manuals.
- Some auto checkweighers may use slightly different procedures. Refer to the individual specifications of your auto checkweigher for details.
- Part of this manual may be changed without prior notice for quality improvement or other reasons.

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Chapter 1 Specification

1.1 Protocol Introduction

Modbus protocol is communication protocol for PLC, developed by Modicon.inc (now Schneider Electric) in United States. Its specifications are publicly available, and is now one of the most commonly be used means of communication in the FA industry. Also, there are three kinds of Modbus protocol, Modbus/RTU, Modbus/ASCII and Modbus/TCP. Modbus/RTU and Modbus/ASCII are used on serial communication such as RS-232C and RS-485. Modbus/TCP is used on TCP/IP communication such as Ethernet. This unit supports only Modbus/TCP.

1.2 Protocol Layout

The Modbus/TCP works by sending a request from the client to the server, and the server returns a response to the client requested.

Each data transports 1byte(8bit) data as it is.

In Modbus/TCP, data structure of the request is the same as that of the response.

Transaction identifier	Protocol identifier	Length	Unit identifier	FUNCTION	Data
Byte 0, 1	Byte 2,3	Byte 4,5	Byte 6	Byte 7	Byte 8...

Transaction identifier	The server returns the same transaction identifier as the requested message. Used for message management by client side.
Protocol identifier	Fixed at 「0」
Length	Number of byte sequences (binary 2 bytes) from [Unit Identifier] (including unit identifier) are sent and received. Byte 4 is the upper binary value and Byte 5 is the lower binary value.
Unit identifier	Fixed at 「1」
FUNCTION	Defined for each function to be executed. See Chapter 3 Technical Specifications (p. 10) for details.
Data	Defined for each function to be executed. See Chapter 3 Technical Specifications (p. 10) for details.

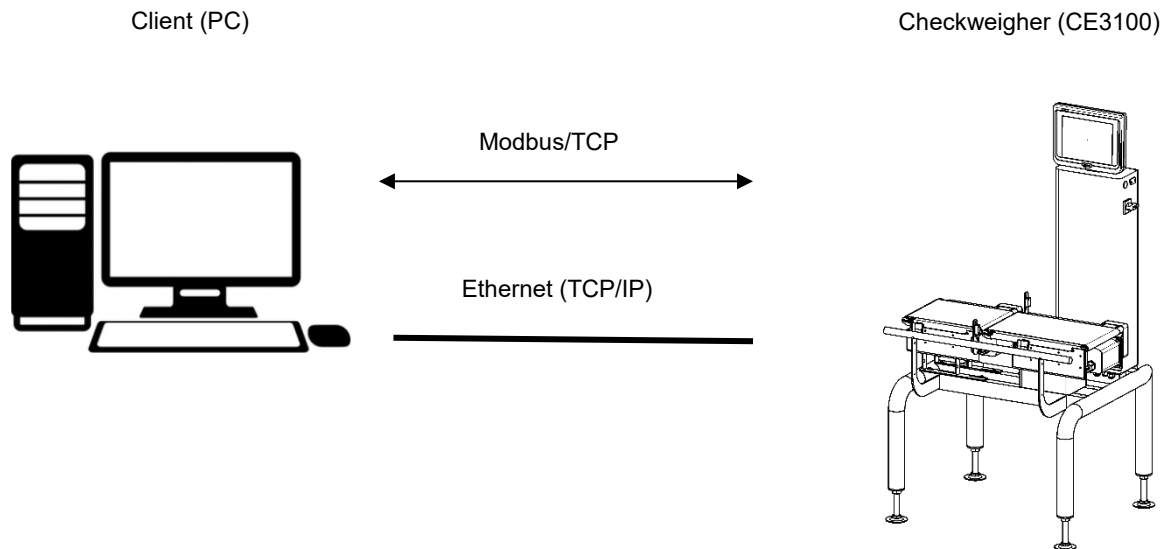
1.3 Exception Response Code

If the response to the requested data is failed, such as if the client does not have the authority to change the data, or if the starting address of the requested data does not correspond to the specified FUNCTION, in "[1.2 Protocol Layout](#)", [FUNCTION] is set to the value obtained by adding [FUNCTION] of the request and 0x80, and the [Data] is set to one byte of the exception response code as shown below.

Code	Contents
1	FUNCTION is not a normal value.
2	The starting address of the requested data does not correspond to the specified FUNCTION.
3	The client does not have the authority to refer to or modify the Data.
4	The process did not complete correctly when responding.

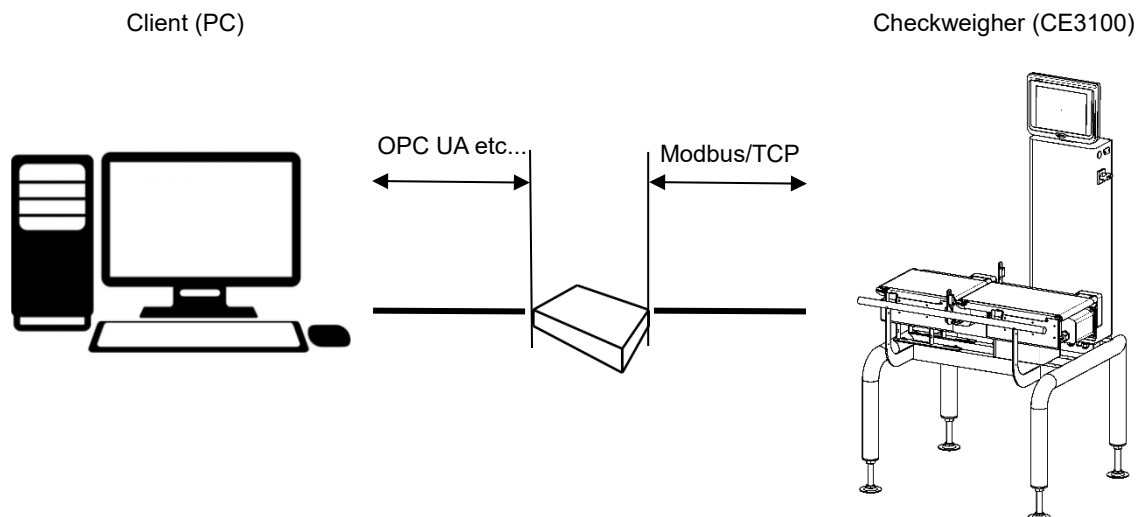
1.4 System Configuration

- Basic System Configuration



Modbus/TCP communication is performed with a PC, PLC, etc. as a client and a Checkweigher as a server.

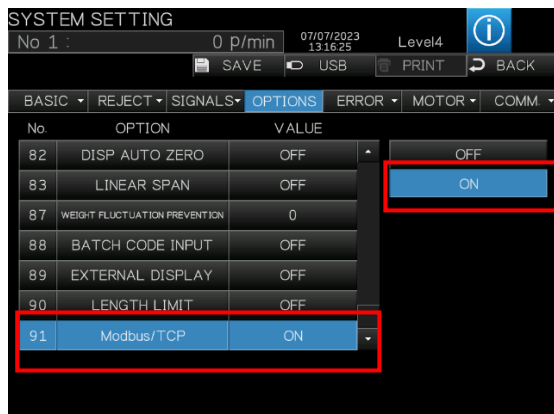
- Other System Configurations (using communication protocol converters)



Modbus/TCP communication is performed between the Checkweigher and the communication protocol converter, with the Checkweigher as the server. The communication protocol converter and the client communicate with each other using OPC UA or other means.

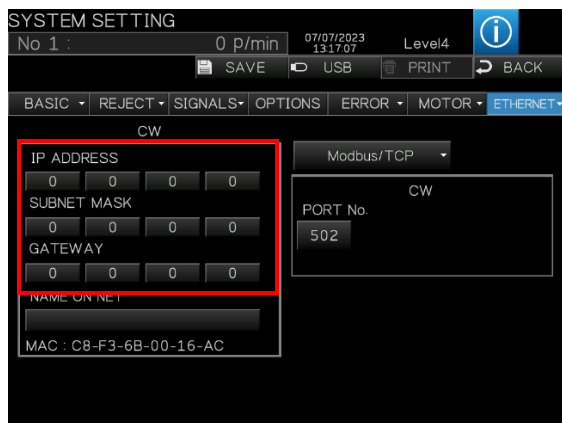
Chapter 2 Functional Specifications

2.1 Setting Modbus/TCP Communication on CE3100(Check Weighter)



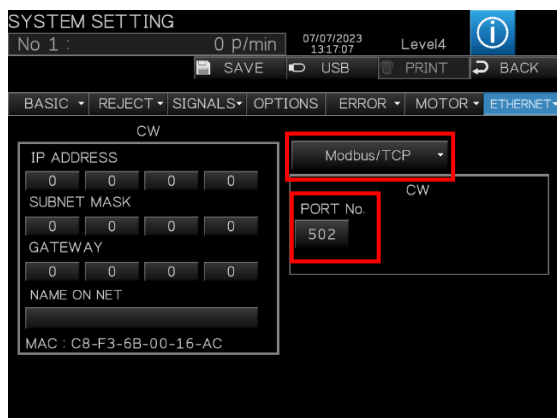
Set [91 Modbus/TCP] to ON on the options Display. (SYSTEM SETTING > OPTION).

caution) This operation needs password input.(paid option)



Setup IP address and Subnet mask. (SYSTEM SETTING > COMMUNICATION > ETHERNET)

caution) If the client is on a different network than the Checkweigher via a router, GATEWAY settings are required..



After setting Modbus/TCP port number by selecting [Modbus/TCP] from list box on the right side on the display, save SYSTEM SETTING and reboot checker.

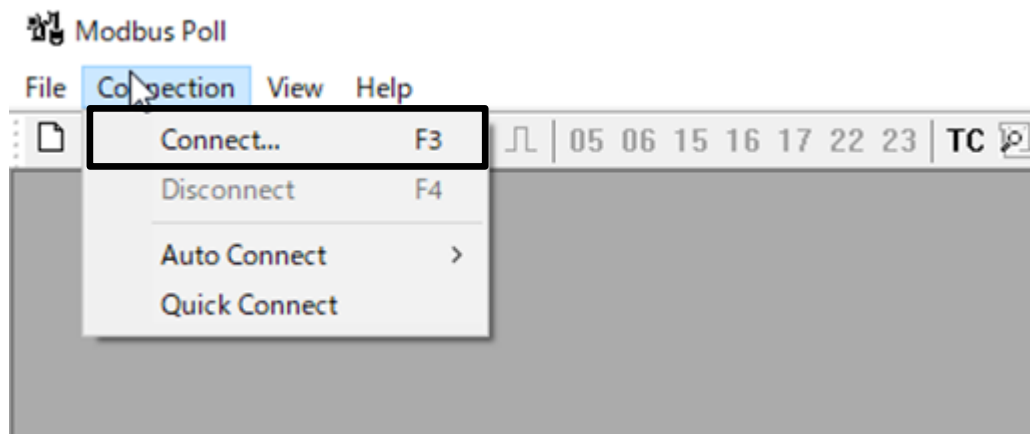
That's all for Modbus/TCP communication setting on the CE3100 side.

2.2 How to use from client side

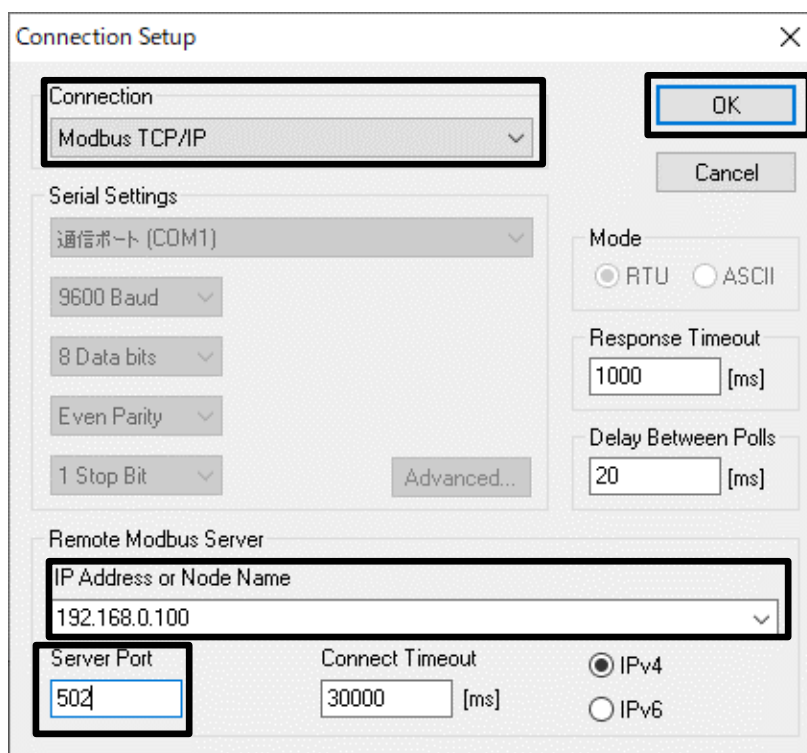
As an example of using Modbus/TCP function of the Checkweigher from a client device, the following shows how to use Modbus Poll to connect to the Checkweigher and send and receive data., to use.

2.2.1 How to connect

Select the [Connection] tab in the upper left corner of the Display and click [Connect...] to display [Connection Setup] screen.



Set [Connection] to [Modbus TCP/IP], and set [IP Address or Node Name] and [Server Port] to the values of [IP Address] and [Port No.] which are set in "[2.1 Setting Modbus/TCP communication on CE3100\(Check Weigher\)](#)" respectively. Click [OK] button to complete the connection with CE3100.

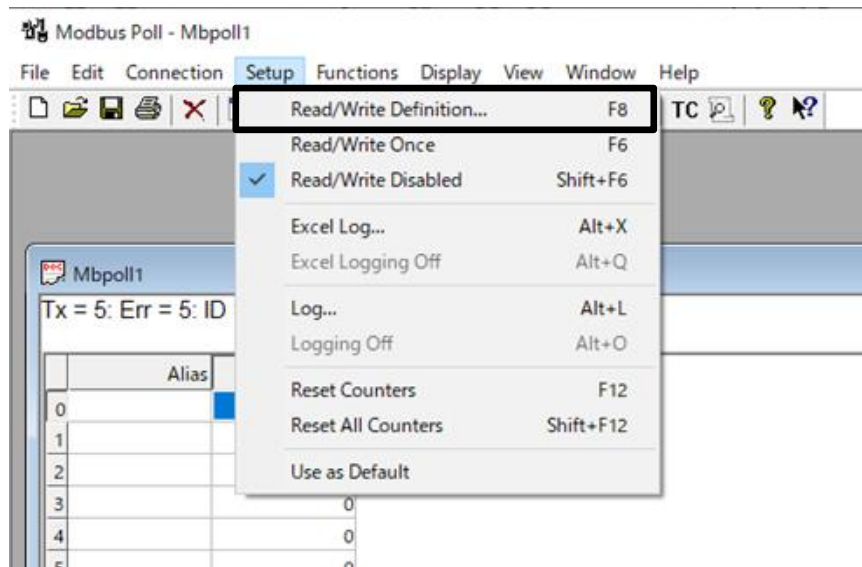


2.2.2 Execution of FUNCTION

Modbus/TCP communication is performed by selecting the function to be executed from several FUNCTION and sending/receiving data.

For an overview of each FUNCTION, refer [Chapter 3 Technical Specifications \(p. 10\)](#).

After the connection is established, select [Setup] tab in the upper left corner of the display and click [Read/Write Definition] and be display [Read/Write Definition] screen.



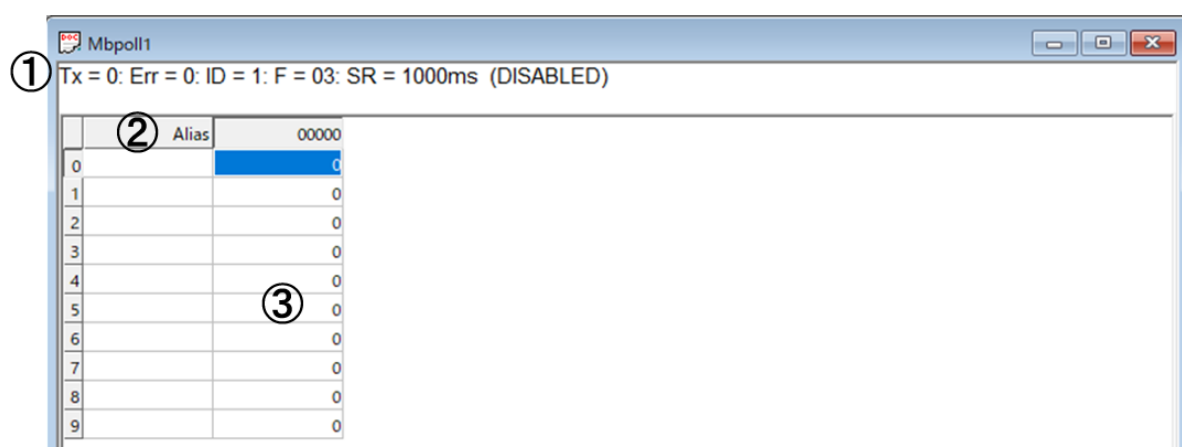
Each of the settings on [Read/Write Definition] screen is as follows.

The screenshot shows the 'Read/Write Definition' dialog box. The settings are as follows:

- ① Slave ID: 1
- ② Function: 03 Read Holding Registers (4x)
- ③ Address: 0
- ④ Quantity: 10
- ⑤ Scan Rate: 1000 [ms]
- ⑥ Read/Write Disabled: ☒ Read/Write Disabled, ☐ Disable on error
- ⑦ View: Rows: ☒ 10, ☐ 20, ☐ 50, ☐ 100, ☐ Fit to Quantity
- ⑧ Read/Write Once: ☐ Read/Write Once
- ⑨ Apply:
- ⑩ OK:
- ⑪ Cancel:

- | | |
|-----------------------|--|
| ① Slave ID | Fixed at [1] on Modbus/TCP communication with CE3100. |
| ② FUNCTION | Set the FUNCTION to be used.
Refer to Chapter 3 Technical Specifications (p. 10) for details. |
| ③ Address | The starting position of the address of the data to be sent/received in the function.
Refer to Chapter 4, Address Map (p. 19) for the address of each data. |
| ④ Quantity | The quantity of the register in which data is to be sent and received by the FUNCTION.
If the function used is FUNCTION 5 or FUNCTION 6, this value is fixed at [1].
The upper limit of the quantity of registers that can be sent and received at one time depends on the function used. For details, refer to each function in Chapter 3, Technical Specifications (p. 10) . |
| ⑤ Scan Rate | The time period for periodic execution of FUNCTION.
To perform cyclic execution, turn off the checkbox ⑥. |
| ⑥ Read/Write Disabled | When the check box is OFF, function is executed at the time period set in ⑤. |
| ⑦ Rows | The display format of the data can be set. |
| ⑧ Read/Write Once | Execute function only once with the current setting information |
| ⑨ Apply | Reflects the setting. |
| ⑩ OK | Reflects the setting, and close [Read/Write Definition] screen. |
| ⑪ Cancel | Discard the setting change status and close [Read/Write Definition] screen. |

After completing the settings on [Read/Write Definition] screen, by entering the F6 key, the data sending/receiving process for the specified address is executed once.
If the function cycle execution is turned on, function is automatically executed according to the set Scan Rate cycle.
The following is an explanation of the notation on the window.



- ① setting information Displays the current setting status.
Each Contents is as follows.
Tx: Number of times sent/received
Err: Number of failures in sending/receiving
ID: Slave ID
F: ID of function
SR: Scan Rate
(DISABLED): Displayed by side [Scan Rate] when cyclic execution of function is OFF.
- ② Alias Contents of each address can be described.
- ③ Data Display The data to be read or written is displayed.
The target address is indicated by a 5-digit number in the column + a number in the row.
The number of columns can be set by Rows in the [Read/Write Definition] screen.
For example, if Rows is 10 and referring the data at address 1048, the value of "1040" in the column and "8" in the row will be the data of 1048 addresses.

In Modbus/TCP communication, one address can contain 2 bytes of data.
Therefore, when sending/receiving 4-byte data, two addresses must be read/written at the same time.
Refer to [3.3 Data Format \(p 17\)](#) for the format according to the type of data to be sent/received.
In Modbus/TCP communication, multiple data at non-consecutive address cannot be read simultaneously.
Therefore, if you want to get the data of address1000 and address1010 at the same time, you need to receive the data of address1000 to address1010 at once.

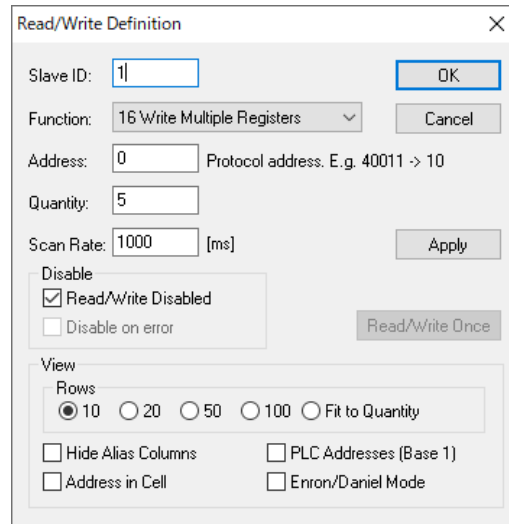
2.2.3 Change of Operating Authority

When executing each function, the operation authority is set for each target register.

If the operating authority of the session is lower than the operating authority set for the target register, the result of executing each function is failure and an exception response code is sent as a response.

(Refer to [1.3 Exception Response Codes \(p 1\)](#))

To change the operating authority, set the following on [Read/Write Definition] screen.



Read/Write Definition

Slave ID: 1 OK Cancel

Function: 16 Write Multiple Registers

Address: 0 Protocol address. E.g. 40011 -> 10

Quantity: 5

Scan Rate: 1000 [ms] Apply

Disable
☒ Read/Write Disabled
☐ Disable on error Read/Write Once

View
Rows
☒ 10 ☐ 20 ☐ 50 ☐ 100 ☐ Fit to Quantity
☐ Hide Alias Columns ☐ PLC Addresses (Base 1)
☐ Address in Cell ☐ Enron/Daniel Mode

Then enter the following in the data display section and send it to CE3100.

Address 0 ASCII code of the quantity of operation authority (1 to 4) you want to change in the upper 8 bits.

Lower 8 bits are fixed at [0].

Address 1~4 Enter the password for the operating authority specified in Address 0 in ASCII code by two characters from the upper bit for each address.

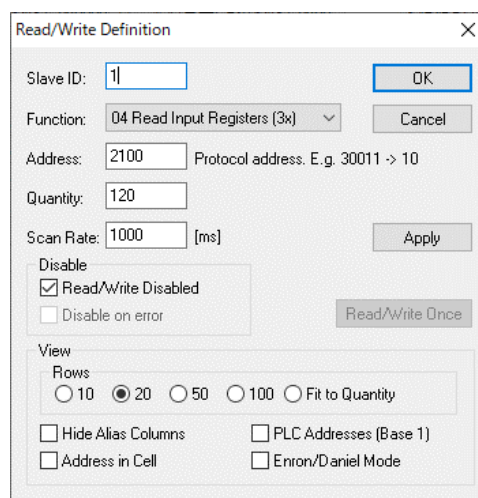
The password for each operating authority is the same as for CE3100.

If the command succeeds, the change in operating authority is complete.

Refer to [Chapter 4, Address Map \(p 19\)](#) for the Referral authority of each register.

2.2.4 Total Data Acquisition

When referring the CE3100's aggregate data via Modbus/TCP communication, configure the following settings on [Read/Write Definition] screen.



Read/Write Definition

Slave ID: 1 OK Cancel

Function: 04 Read Input Registers (3x)

Address: 2100 Protocol address. E.g. 30011 -> 10

Quantity: 120

Scan Rate: 1000 [ms] Apply

Disable
☒ Read/Write Disabled
☐ Disable on error Read/Write Once

View
Rows
☐ 10 ☒ 20 ☐ 50 ☐ 100 ☐ Fit to Quantity
☐ Hide Alias Columns ☐ PLC Addresses (Base 1)
☐ Address in Cell ☐ Enron/Daniel Mode

With this setting, the total data allocated to addresses between 2100 and 2219 are received in one batch.

Refer to [4.4 Input Register \(P.35\)](#) for the data contents of each address.

2.2.5 Acquisition of Weighing Data

The last 100 weighing data can be read over Modbus/TCP communication.
Set the following in [Read/Write Definition] screen.

Read/Write Definition

Slave ID: 1

Function: 04 Read Input Registers (3x)

Address: 3000 Protocol address. E.g. 30011 -> 10

Quantity: 22

Scan Rate: 1000 [ms]

Disable

☒ Read/Write Disabled

☐ Disable on error

Read/Write Once

View

Rows

☒ 10 ☐ 20 ☐ 50 ☐ 100 ☐ Fit to Quantity

☐ Hide Alias Columns ☐ PLC Addresses (Base 1)

☐ Address in Cell ☐ Enron/Daniel Mode

With this setting, the weighing data assigned to addresses between 3000 and 3021 is received in a batch. Each time the command is sent once, the oldest weighing data held inside the CE3100 can be read. The next weighing data can be read by executing the same function again.
Refer to [4.4 Input Register \(P.35\)](#) for the data Contents of each address.
Since individual data reception is not supported for weighing data, be sure to receive all data in one batch.

2.2.6 Refer to System Settings and Program Settings

To get information on the CE3100's system settings and Program Settings via Modbus/TCP communication, specify the address and number of the data to be gotten and execute function in the same way as described in [2.2.4 Total Data Acquisition \(P.8\)](#).

Refer to [4.3 Holding Register \(P.20\)](#) and [4.4 Input Register \(P.35\)](#) for the data Contents of each address.

2.2.7 Changing System Settings and Program Settings

To edit the CE3100's system setting and Program setting via Modbus/TCP communication, the edit state must be changed first.

Use function 5 or 15 to change the value of the Edit Flag for the System Setting or Program Setting to 1
Refer to [4.1 Coil \(p. 19\)](#) for the target address of each flag.

When the Program Setting Edit Flag is set to 1, the Program Setting can be changed, and when System Setting Edit Flag is set to 1, the system setting can be changed.

In this state, the address corresponding to the item can be specified and the setting value can be changed. To reflect the changed values in the CE3100, change the values of [Program Setting Edit Flag] and [System Setting Edit Flag] to 0.

Chapter 3 Technical Specifications

3.1 About Specifics of Modbus Protocol

The Modbus protocol specifications are described in the publicly available protocol specification (PI-MDBS-300Rev J), please refer to this specification for detailed Modbus protocol specifications.

3.2 Each Function Information

This section shows the data structures and corresponding functions of the [Request] and [Response] message portions of various FUNCTIONS.

The [Field Length] of the [Response] in each FUNCTION varies according to the [Number of Registers] in the [Request] unless otherwise stated.

The data structure of the [Response] shown here are all based on the case of successful execution of the requested FUNCTION.

Refer to [1.2 Protocol Layout \(p. 1\)](#) and [1.3 Exception Response Code \(p. 1\)](#) for the data structure in the case of a failed FUNCTION execution.

3.2.1 FUNCTION 1 Read Coil Status

This FUNCTION refers to binary data (Coil) that can be referenced and changed.

The address range of Coil is 0 to 9999.

Byte	Request		Response	
	Field Name	Value	Field Name	Value
0	Transaction	0x00	Transaction	0x00
1		0x01		0x01
2	Protocol ID	0x00	Protocol ID	0x00
3		0x00		0x00
4	Field Length	0x00	Field Length	
5		0x06		
6	Unit Identifier	0x01	Unit Identifier	0x01
7	Function Code	0x01	Function Code	0x01
8	Initial Address		Number of Data Bytes	
9			Data	
10	Number of Registers		:	
11			:	

From the [Data] of the [Response], the data with quantity specified in [Number of Registers] is generated from the address specified in [Initial Address] of [Request]

The value that can be specified for the [Number of Registers] is from 1 to 2000.

The [Data] part stores the value of each address (1 for ON, 0 for OFF) in order from the lower bits, and one byte (8 bits) represents the ON/OFF of 8 registers.

As an example, if the [Initial Address] of the [Request] is 0x00, 0x03 and the [Number of Registers] is 0x00, 0x0A, the request range is data corresponding to addresses 3 to 12.

At this time, the [Number of Data Bytes] in the [Response] is 2.

Therefore, if the data corresponding to addresses 3 to 12 are

ON - OFF - ON - ON - OFF - ON - OFF - ON - ON - OFF, the [Data] in the [Response] will be 0b10101101 (0xAD) and 0b00000001 (0x01) in binary.

※If the number of registers to be stored in the data is less than 8, the remaining upper bits are filled with 0.

3.2.2 FUNCTION 2 Read Input Status

This FUNCTION refers to binary data (Input Status) that cannot be changed, only referred to.

The range of addresses actually used for Input Status is 10000 to 19999, but the [Initial Address] of [Request] is specified as 0 to 9999, omitting the most significant digit.

※Specifications may differ depending on the client software.

Byte	Request		Response	
	Field Name	ex.	Field Name	ex.
0	Transaction	0x00	Transaction	0x00
1		0x01		0x01
2	Protocol ID	0x00	Protocol ID	0x00
3		0x00		0x00
4	Field Length	0x00	Field Length	
5		0x06		
6	Unit Identifier	0x01	Unit Identifier	0x01
7	Function Code	0x02	Function Code	0x02
8	Initial Address		Number of Data Bytes	
9			Data	
10	Number of Registers		:	
11			:	

The [Data] in the [Response] and there after generate the number of Input Status data specified in the [Number of Registers] starting from the [Initial Address] of the [Request] in sequence.

The value that can be specified for the [Number of Registers] is from 1 to 2000.

The [Data] portion of the [Response] stores the value of each address of the target (1 for ON, 0 for OFF) in order from the lower bits as in [3.2.1 FUNCTION 1 Read Coil Status \(p. 10\)](#), and 1 byte (8 bits) represents ON/OFF of 8 registers.

※If the number of registers to be stored in the data is less than 8, the remaining upper bits are filled with 0.

3.2.3 FUNCTION 3 Read Holding Register

This FUNCTION refers to 2-byte length data (Holding Register) that can be referred to and changed. The range of addresses actually used for the Holding Register is 30000 to 39999, but the [Initial Address] of [Request] is specified as 0 to 9999, omitting the most significant digit.

※Specifications may differ depending on the client software.

Byte	Request		Response	
	Field Name	Ex	Field Name	Ex.
0	Transaction	0x00	Transaction	0x00
1		0x01		0x01
2	Protocol ID	0x00	Protocol ID	0x00
3		0x00		0x00
4	Field Length	0x00	Field Length	
5		0x06		
6	Unit Identifier	0x01	Unit Identifier	0x01
7	Function Code	0x03	Function Code	0x03
8	Initial Address		Number of Data	
9			Data	
10	Number of Registers			
11			:	

Enter the number of Holding Registers to be referenced in the [Number of Registers] field under [Request]. Because the Holding Register is 2 bytes long, for example, 2 registers are required to reference 4 bytes of data.

The value that can be entered is from 1 to 125.

After the [Data] of the [Response], the Holding Register for the [Number of Registers] is stored in 2-byte units, starting from the [Initial Address] of the [Request].

The [Number of Data Bytes] of the [Response] contains the total number of bytes of the Holding Register stored in the [Data] of the [Response].

3.2.4 FUNCTION 4 Read Input Register

This FUNCTION refers to 2-byte length data (Input Register) that cannot be changed, only referred to. The range of addresses actually used for the Input Register is 40000 to 49999, but the [Initial Address] of [Request] is specified as 0 to 9999, omitting the most significant digit.

※Specifications may differ depending on the client software.

Byte	Request		Response	
	Field Name	ex.	Field Name	ex.
0	Transaction	0x00	Transaction	0x00
1		0x01		0x01
2	Protocol ID	0x00	Protocol ID	0x00
3		0x00		0x00
4	Field Length	0x00	Field Length	
5		0x06		
6	Unit Identifier	0x01	Unit Identifier	0x01
7	Function Code	0x04	Function Code	0x04
8	Initial Address		Number of Data Bytes	
9			Data	
10	Number of Registers			
11			:	

Enter the number of Input Registers to be referred to in the [Number of Registers] field under [Request]. Since the Input Register is 2 bytes long, for example, 2 registers are required to refer to 4 bytes of data. The value that can be entered is from 1 to 125.

After the [Data] in the [Response], the Input Register for the [Number of Registers] is stored in 2-byte units, starting from the [Initial Address] of the [Request].

The [Number of Data Bytes] of the [Response] contains the total number of bytes of Input Register stored in the [Data] of the [Response].

3.2.5 FUNCTION 5 Force Single Coil

This FUNCTION changes the Coil data.

Note that this FUNCTION can only change a single piece of data. If you want to change multiple coil data at once, use [3.2.7 FUNCTION 15 Force Multiple Coils \(p. 15\)](#).

Byte	Request		Response	
	Field Name	ex.	Field Name	ex.
0	Transaction	0x00	Transaction	0x00
1		0x01		0x01
2	Protocol ID	0x00	Protocol ID	0x00
3		0x00		0x00
4	Field Length	0x00	Field length	0x00
5		0x06		0x06
6	Unit Identifier	0x01	Unit Identifier	0x01
7	Function Code	0x05	Function Code	0x05
8	Address		Address	
9				
10	Changed Data		Changed Data	
11				

Changes the state of the data specified by [Address] to either ON or OFF.

[Changed Data] is 0xFF, 0x00 when ON, and 0x00, 0x00 when OFF.

The [Response] will be the same message as the [Request].

3.2.6 FUNCTION 6 Preset Single Register

This FUNCTION changes the Holding Register data. It can only change 2bytes data. To change multiple Holding Register data at once, or to change 4 or 8 bytes of Holding Register data, use [3.2.8 FUNCTION 16 Preset Multiple Registers \(P.16\)](#).

Byte	Request		Response	
	Field Name	ex.	Field Name	ex.
0	Transaction	0x00	Transaction	0x00
1		0x01		0x01
2	Protocol ID	0x00	Protocol ID	0x00
3		0x00		0x00
4	Field Length	0x00	Field Length	0x00
5		0x06		0x06
6	Unit Identifier	0x01	Unit Identifier	0x01
7	Function Code	0x06	Function Code	0x06
8	Address		Address	
9				
10	Changed Data		Changed Data	
11				

Specify the [Address] to be changed and the [Changed Data] in 2-byte.
The [Response] message is the same as the [Request].

3.2.7 FUNCTION 15 Force Multiple Coils

This FUNCTION changes Coil data.

The Coil data of the specified [Number of Registers] can be changed at once.

Byte	Request		Response	
	Field Name	ex.	Field Name	ex.
0	Transaction	0x00	Transaction	0x00
1		0x01		0x01
2	Protocol ID	0x00	Protocol ID	0x00
3		0x00		0x00
4	Field Length		Field Length	0x00
5				0x06
6	Unit Identifier	0x01	Unit Identifier	0x01
7	Function Code	0x0F	Function Code	0x0F
8	Initial Address		Initial Address	
9				
10	Number of Registers		Number of Data Bytes	
11				
12	Number of Data Bytes			
13	Changed Data			
14				
15	:			
~				

Changes the state of the data specified by the [Initial Address] and [Number of Registers] to either ON or OFF.

The value that can be specified for the [Number of Registers] is from 1 to 2000.

The [Changed Data] part stores the value of each address (1 for ON, 0 for OFF) in order from the lower bits, and one byte (8 bits) represents the ON/OFF status of 8 registers.

As an example, if the [Initial Address] of [Request] is 0x00, 0x03 and the [Number of Registers] is 0x00, 0x0A, the request range is data corresponding to addresses 3 to 12.

At this time, the [Number of Data Bytes] for [Request] and [Response] is 2.

Therefore, the data corresponding to addresses 3 to 12 will be changed to ON - OFF - ON - ON - OFF - ON - OFF - ON - ON - OFF respectively, the [Changed Data] in [Request] will be 0b10101101 (0xAD) and 0b00000001 (0x01) in binary.

※If the number of registers to be stored in the data is less than 8, the remaining upper bits are filled with 0.

3.2.8 FUNCTION 16 Preset Multiple Registers

This FUNCTION changes the Holding Register data.

Holding Registers with quantity specified in [Number of Register] can be changed at once.

Byte	Request		Response	
	Field Name	ex.	Field Name	ex.
0	Transaction	0x00	Transaction	0x00
1		0x01		0x01
2	Protocol ID	0x00	Protocol ID	0x00
3		0x00		0x00
4	Field Length		Field Length	0x00
5				0x06
6	Unit Identifier	0x01	Unit Identifier	0x01
7	Function Code	0x10	Function Code	0x10
8	Initial Address		Initial Address	
9				
10	Number of Registers		Number of Registers	
11				
12	Number of Data Bytes			
13	Changed Data			
14				
15 ~	:			

Change the data specified in [Initial Address] and [Number of Registers] in [Request].

Enter the number of Holding Registers to be referenced in the [Number of Registers] field.

Since the Holding Register is 2 bytes long, for example, 2 registers are required to reference 4 bytes of data.

The value that can be entered is from 1 to 125.

After [Changed Data] of [Request], set 2 bytes in hexadecimal for each target data in order from [Initial Address] of [Request].

Set the total number of bytes in the Input Register to be stored in the [Changed Data] of the [Request] to the [Number of Data Bytes] of the [Request].

The [Initial Address] and [Number of Registers] in the [Response] will contain the same values as in the [Request].

3.3 Data Format

The data format for each data type is as follows.

When sending and receiving multiple array data at once, the data with element number 0 is stored at the corresponding address in order.

※When sending and receiving data as a character string, set the trailing data to be 0x00.

For addresses that handle data as character strings, refer to "[Data Type] *00(number of characters)" in [Type] in [Chapter 4, "Address Map" \(p. 19\)](#).

3.3.1 Data Type: CHAR (C) 2 Bytes

When sending and receiving CHAR type data, two pieces of data are converted into one 2-byte piece of data and exchanged.

Ex.) When sending and receiving two pieces of data
(Byte0, Byte1 [element number 1])

ADDRESS+0
Upper 8bit: Byte 0
Lower 8bit: Byte 1

When sending and receiving only one piece of data (Byte 0)

ADDRESS+0
Upper 8bit: Byte0
Lower 8bit: 0x00

3.3.2 Data Type: SHORT(S) 2 Bytes

Data of type SHORT is transmitted and received with one data assigned to one address.

Ex.)

ADDRESS+0
Data

3.3.3 Data Type: LONG (L) 4 Bytes

LONG type data uses two addresses to send and receive one piece of data.

Sending and receiving are performed with the upper 2 bytes and lower 2 bytes of LONG type data assigned to each address.

Ex.) Original data MSB [Data1] [Data0] LSB

ADDRESS+0	+1
Data1	Data0

3.3.4 Data Type: LONGLONG (LL) 8 Bytes

LONGLONG type data uses four addresses to send and receive one piece of data.

Sending and receiving are performed with the most significant 2 bytes, the most significant 2 bytes, the least significant 2 bytes, and the most significant 2 bytes of LONGLONG type data allocated to each address.

Ex.) Original data MSB [Data3] [Data2] [Data1] [Data0] LSB

ADDRESS+0	+1	+2	+3
Data3	Data2	Data1	Data0

3.3.5 Data type: DOUBLE (D) 8 bytes

DOUBLE type data uses four addresses to send and receive one piece of data.

sending and receiving are performed with the sign, exponent part, and mantissa part of the DOUBLE type data assigned to each address.

Ex) In the case of sending and receiving one DOUBLE type data

ADDRESS+0				+1				+2				+3			
15	14	...	4	3	...	0	15 ... 0	15 ... 0	15 ... 0	15 ... 0	15 ... 0	15 ... 0	15 ... 0	15 ... 0	15 ... 0
sign				exponent (11bit)				mantissa part (52bit)							

$$\text{Value} = (-1)^{\text{sign part}} \times 2^{\text{exponent} - 1023} \times (1 + \text{mantissa} \times 2^{-52})$$

Chapter 4 Address Map

The address map may be updated without notice.

When an address not listed in the address map is referred to or edited as a [Initial Address], an exception response code is sent.

When an address marked as [NONE] in the address map is referred to as the [Initial Address] or edited, 0 is sent. (No exception response code is sent.)

[Referral Authority] in the table is the operating authority required to read the value of the address.

[Editorial Authority] in the table indicates the operating authority required to write values to the address.

Refer to [2.2.3 Change Operating Authority \(p. 8\)](#) for how to change the operating authority.

4.1 Coil

ADDRESS	Contents	Bit return		Referral Authority	Editorial Authority
		0	1		
1000	Clear Total	-	DO	ALL(※1)	Level2~ (※2)
1001	Switch Operation Status	STOP	START	ALL(※1)	ALL
1002	Adjust Zero Point	-	DO	ALL(※1)	ALL (※2)
1003	Program Setting Edit Flag	Unreferred	Referring	Level2~	Level2~
1004	System Setting Edit Flag	Unreferred	Referring	Level2~	Level2~

(※1) When reading the value of each ADDRESS, 0 is sent.

(※2) After the execution of FUNCTION is completed, the value of the corresponding ADDRESS changes to 0.

4.2 Input Status

ADDRESS is originally from 10000 to 19999, but the highest level is omitted from the description.

ADDRESS	Contents	Bit return		Referral Authority
		0	1	
0000	CW Operation Status Acquisition	STATIC	In Operation	ALL

4.3 Holding Register

ADDRESS is originally from 30000 to 39999, but the highest level is omitted from the description.

ADDRESS	Contents	Type	Byte	Ref.Auth	Edi.Auth
Present Program Number					
0005	Present Program number	S	2	All	All
System Setting - Basic Setting					
4000	MODEL NAME	C*8	16	Level4	Level4
4008	SERIAL NO.	C*8	16	Level4	Level4
System Setting - Measurement					
4100	DECIMAL POINT	S	2	Level4	Level4
4101	INCREMENT	S	2	Level4	Level4
4102	UNIT	S	2	Level4	Level4
4103	T1-T2 %	S	2	Level4	Level4
4104	HIGH RESOLUTION	S	2	Level4	Level4
4105	MULTI INTERVAL	S	2	Level4	Level4
4106	BOUND/ARY1	L	4	Level4	Level4
4108	BOUND/ARY2	L	4	Level4	Level4
4110	VERIFICATION MODE (ON/OFF)	S	2	Level4	Level4
4111	ACCURACY CLASS	S	2	Level4	Level4
4112	DEC. POSITION (ACTUAL SCALE INTERVAL)	S	2	Level4	Level4
4113	INCREMENT (ACTUAL SCALE INTERVAL)	S	2	Level4	Level4
4114	NONE	L	4	Level4	Level4
4116	NONE	L	4	Level4	Level4
4118	DEC. POSITION (GRADE DESIGNATION)	S	2	Level4	Level4
4119	INCREMENT(GRADEDESIGNATION)	S	2	Level4	Level4
4120	NONE	S	2	Level4	Level4
4121	DEC.POSITION(INDICATIONFOR VERIFICATION)	S	2	Level4	Level4
4122	INCREMENT (INDICATIONFOR VERIFICATION)	S	2	Level4	Level4
4123	CATCHWEIGHER TYPE	S	2	Level4	Level4
4124	DEC. POSITION (VERIFICATION SCALE1)	S	2	Level4	Level4
4125	INCREMENT (VERIFICATION SCALE1)	S	2	Level4	Level4
4126	DEC. POSITION (VERIFICATION SCALE2)	S	2	Level4	Level4
4127	INCREMENT (VERIFICATION SCALE2)	S	2	Level4	Level4
4128	CHANGING POINT	L	4	Level4	Level4
4130	NONE	S	2	Level4	Level4
⋮				Level4	Level4
4160	NONE	S	2	Level4	Level4
System Setting - Weighing Conveyor, Load Cell					
4200	CONVEYOR TYPE	S	2	Level4	Level4
4201	CONVEYOR LENGTH	L	4	Level4	Level4
4203	SCALE	L	4	Level4	Level4
4205	SCALE OVER	L	4	Level4	Level4
4207	ACCURACY	L	4	Level4	Level4
4209	INITIAL ZERO RANGE	L	4	Level4	Level4
4211	ZERO RANGE	L	4	Level4	Level4
4213	LOWEST LIMIT	L	4	Level4	Level4
4215	STABILITY RANGE	L	4	Level4	Level4

ADDRESS	Contents	Type	Byte	Ref Auth	Edi Auth
4217	NONE	S	2	Level4	Level4
:				Level4	Level4
4229	NONE	S	2	Level4	Level4
System Setting - Static Filter					
4300	STATIC FILTER1(SF1)	S	2	Level4	Level4
4301	STATIC FILTER2(SF2)	S	2	Level4	Level4
4302	STATIC FILTER3(SF3)	S	2	Level4	Level4
4303	STATIC FILTER4(SF4)	S	2	Level4	Level4
4304	SF AVERAGE COUNT	S	2	Level4	Level4
4305	D.ZERO ADJ AVG. COUNT	S	2	Level4	Level4
4306	SF MEAS STABILITY RANGE	S	2	Level4	Level4
System Setting - Display					
4400	INSTALLED LANGUAGE	L	4	Level4	Level4
4402	DATE	S	2	Level4	Level4
4403	WARM UP LENGTH	S	2	Level4	Level4
4404	LIMITATION RANGE IN AUTO-WEIGHT(UNIT)	S	2	Level4	Level4
4405	LIMITATION RANGE IN AUTO-WEIGHT(HI LIMIT)	S	2	Level4	Level4
4406	LIMITATION RANGE IN AUTO-WEIGHT(LOLIMIT)	S	2	Level4	Level4
System Setting - Login					
4500	LOGIN METHOD	S	2	Level4	Level4
System Setting - Motor					
4600	MOTOR 1 MOTOR ALLOCATION	S	2	Level4	Level4
4601	MOTOR1 OPTION MOTOR	S	2	Level4	Level4
4602	MOTOR1 MAX SPEED	S	2	Level4	Level4
4603	MOTOR1 MAX SPEED D/A	S	2	Level4	Level4
4604	MOTOR1 MIN SPEED	S	2	Level4	Level4
4605	MOTOR1 MIN SPEED D/A	S	2	Level4	Level4
4606	NONE	L	4	Level4	Level4
4608	NONE	L	4	Level4	Level4
4610	MOTOR1 MT DIVERter LOGIC	S	2	Level4	Level4
4611	MOTOR1 GEAR HEAD	S	2	Level4	Level4
4612	MOTOR1 PULLEY DIAMETER	S	2	Level4	Level4
4613	MOTOR1 CONVEYOR GEAR TOOTH	S	2	Level4	Level4
4614	MOTOR1 MT PULLEY GEAR TOOTH	S	2	Level4	Level4
4615	MOTOR1 DIRECTION	S	2	Level4	Level4
4616	MOTOR1 PLS COUNT	S	2	Level4	Level4
4617	MOTOR1 ERROR COEFFICIENT	S	2	Level4	Level4
4618	MOTOR1 DETECT ALARM OUTPUT	S	2	Level4	Level4
4619	MOTOR2 MOTOR ALLOCATION	S	2	Level4	Level4
4620	MOTOR2 OPTION MOTOR	S	2	Level4	Level4
4621	MOTOR2 MAX SPEED	S	2	Level4	Level4
4622	MOTOR2 MAX SPEED D/A	S	2	Level4	Level4
4623	MOTOR2 MIN SPEED	S	2	Level4	Level4
4624	MOTOR2 MIN SPEED D/A	S	2	Level4	Level4
4625	NONE	L	4	Level4	Level4
4627	NONE	L	4	Level4	Level4
4629	MOTOR2 MT DIVERter LOGIC	S	2	Level4	Level4
4630	MOTOR2 GEAR HEAD	S	2	Level4	Level4

ADDRESS	Contents	Type	Byte	Ref. Auth	Edi. Auth
4631	MOTOR2 PULLEY DIAMETER	S	2	Level4	Level4
4632	MOTOR2 CONVEYOR GEAR TOOTH	S	2	Level4	Level4
4633	MOTOR2 MT PULLEY GEAR TOOTH	S	2	Level4	Level4
4634	MOTOR2 DIRECTION	S	2	Level4	Level4
4635	MOTOR2 PLS COUNT	S	2	Level4	Level4
4636	MOTOR2 ERROR COEFFICIENT	S	2	Level4	Level4
4637	MOTOR2 DETECT ALARM OUTPUT	S	2	Level4	Level4
4638	MOTOR3 MOTOR ALLOCATION	S	2	Level4	Level4
4639	MOTOR3 OPTION MOTOR	S	2	Level4	Level4
4640	MOTOR3 MAX SPEED	S	2	Level4	Level4
4641	MOTOR3 MAX SPEED D/A	S	2	Level4	Level4
4642	MOTOR3 MIN SPEED	S	2	Level4	Level4
4643	MOTOR3 MIN SPEED D/A	S	2	Level4	Level4
4644	NONE	L	4	Level4	Level4
4646	NONE	L	4	Level4	Level4
4648	MOTOR3 MT DIVERTER LOGIC	S	2	Level4	Level4
4649	MOTOR3 GEAR HEAD	S	2	Level4	Level4
4650	MOTOR3 PULLEY DIAMETER	S	2	Level4	Level4
4651	MOTOR3 CONVEYOR GEAR TOOTH	S	2	Level4	Level4
4652	MOTOR3 MT PULLEY GEAR TOOTH	S	2	Level4	Level4
4653	MOTOR3 DIRECTION	S	2	Level4	Level4
4654	MOTOR3 PLS COUNT	S	2	Level4	Level4
4655	MOTOR3 ERROR COEFFICIENT	S	2	Level4	Level4
4656	MOTOR3 DETECT ALARM OUTPUT	S	2	Level4	Level4
4657	MOTOR4 MOTOR ALLOCATION	S	2	Level4	Level4
4658	MOTOR4 OPTION MOTOR	S	2	Level4	Level4
4659	MOTOR4 MAX SPEED	S	2	Level4	Level4
4660	MOTOR4 MAX SPEED D/A	S	2	Level4	Level4
4661	MOTOR4 MIN SPEED	S	2	Level4	Level4
4662	MOTOR4 MIN SPEED D/A	S	2	Level4	Level4
4663	NONE	L	4	Level4	Level4
4665	NONE	L	4	Level4	Level4
4667	MOTOR4 MT DIVERTER LOGIC	S	2	Level4	Level4
4668	MOTOR4 GEAR HEAD	S	2	Level4	Level4
4669	MOTOR4 PULLEY DIAMETER	S	2	Level4	Level4
4670	MOTOR4 CONVEYOR GEAR TOOTH	S	2	Level4	Level4
4671	MOTOR4 MT PULLEY GEAR TOOTH	S	2	Level4	Level4
4672	MOTOR4 DIRECTION	S	2	Level4	Level4
4673	MOTOR4 PLS COUNT	S	2	Level4	Level4
4674	MOTOR4 ERROR COEFFICIENT	S	2	Level4	Level4
4675	MOTOR4 DETECT ALARM OUTPUT	S	2	Level4	Level4
4676	MOTOR5 MOTOR ALLOCATION	S	2	Level4	Level4
4677	MOTOR5 OPTION MOTOR	S	2	Level4	Level4
4678	MOTOR5 MAX SPEED	S	2	Level4	Level4
4679	MOTOR5 MAX SPEED D/A	S	2	Level4	Level4
4680	MOTOR5 MIN SPEED	S	2	Level4	Level4
4681	MOTOR5 MIN SPEED D/A	S	2	Level4	Level4

ADDRESS	Contents	Type	Byte	Ref. Auth	Edi. Auth
4682	NONE	L	4	Level4	Level4
4684	NONE	L	4	Level4	Level4
4686	MOTOR5 MT DIVERTER LOGIC	S	2	Level4	Level4
4687	MOTOR5 GEAR HEAD	S	2	Level4	Level4
4688	MOTOR5 PULLEY DIAMETER	S	2	Level4	Level4
4689	MOTOR5 CONVEYOR GEAR TOOTH	S	2	Level4	Level4
4690	MOTOR5 MT PULLEY GEAR TOOTH	S	2	Level4	Level4
4691	MOTOR5 DIRECTION	S	2	Level4	Level4
4692	MOTOR5 PLS COUNT	S	2	Level4	Level4
4693	MOTOR5 ERROR COEFFICIENT	S	2	Level4	Level4
4694	MOTOR5 DETECT ALARM OUTPUT	S	2	Level4	Level4
4695	MOTOR6 MOTOR ALLOCATION	S	2	Level4	Level4
4696	MOTOR6 OPTION MOTOR	S	2	Level4	Level4
4697	MOTOR6 MAX SPEED	S	2	Level4	Level4
4698	MOTOR6 MAX SPEED D/A	S	2	Level4	Level4
4699	MOTOR6 MIN SPEED	S	2	Level4	Level4
4700	MOTOR6 MIN SPEED D/A	S	2	Level4	Level4
4701	NONE	L	4	Level4	Level4
4703	NONE	L	4	Level4	Level4
4705	MOTOR6 MT DIVERTER LOGIC	S	2	Level4	Level4
4706	MOTOR6 GEAR HEAD	S	2	Level4	Level4
4707	MOTOR6 PULLEY DIAMETER	S	2	Level4	Level4
4708	MOTOR6 CONVEYOR GEAR TOOTH	S	2	Level4	Level4
4709	MOTOR6 MT PULLEY GEAR TOOTH	S	2	Level4	Level4
4710	MOTOR6 DIRECTION	S	2	Level4	Level4
4711	MOTOR6 PLS COUNT	S	2	Level4	Level4
4712	MOTOR6 ERROR COEFFICIENT	S	2	Level4	Level4
4713	MOTOR6 DETECT ALARM OUTPUT	S	2	Level4	Level4
4714	MOTOR7 MOTOR ALLOCATION	S	2	Level4	Level4
4715	MOTOR7 OPTION MOTOR	S	2	Level4	Level4
4716	MOTOR7 MAX SPEED	S	2	Level4	Level4
4717	MOTOR7 MAX SPEED D/A	S	2	Level4	Level4
4718	MOTOR7 MIN SPEED	S	2	Level4	Level4
4719	MOTOR7 MIN SPEED D/A	S	2	Level4	Level4
4720	NONE	L	4	Level4	Level4
4722	NONE	L	4	Level4	Level4
4724	MOTOR7 MT DIVERTER LOGIC	S	2	Level4	Level4
4725	MOTOR7 GEAR HEAD	S	2	Level4	Level4
4726	MOTOR7 PULLEY DIAMETER	S	2	Level4	Level4
4727	MOTOR7 CONVEYOR GEAR TOOTH	S	2	Level4	Level4
4728	MOTOR7 MT PULLEY GEAR TOOTH	S	2	Level4	Level4
4729	MOTOR7 DIRECTION	S	2	Level4	Level4
4730	MOTOR7 PLS COUNT	S	2	Level4	Level4
4731	MOTOR7 ERROR COEFFICIENT	S	2	Level4	Level4
4732	MOTOR7 DETECT ALARM OUTPUT	S	2	Level4	Level4
4733	MOTOR8 MOTOR ALLOCATION	S	2	Level4	Level4
4734	MOTOR8 OPTION MOTOR	S	2	Level4	Level4
4735	MOTOR8 MAX SPEED	S	2	Level4	Level4

ADDRESS	Contents	Type	Byte	Ref. Auth	Edi. Auth
4736	MOTOR8 MAX SPEED D/A	S	2	Level4	Level4
4737	MOTOR8 MIN SPEED	S	2	Level4	Level4
4738	MOTOR8 MIN SPEED D/A	S	2	Level4	Level4
4739	NONE	L	4	Level4	Level4
4741	NONE	L	4	Level4	Level4
4743	MOTOR8 MT DIVERter LOGIC	S	2	Level4	Level4
4744	MOTOR8 GEAR HEAD	S	2	Level4	Level4
4745	MOTOR8 PULLEY DIAMETER	S	2	Level4	Level4
4746	MOTOR8 CONVEYOR GEAR TOOTH	S	2	Level4	Level4
4747	MOTOR8 MT PULLEY GEAR TOOTH	S	2	Level4	Level4
4748	MOTOR8 DIRECTION	S	2	Level4	Level4
4749	MOTOR8 PLS COUNT	S	2	Level4	Level4
4750	MOTOR8 ERROR COEFFICIENT	S	2	Level4	Level4
4751	MOTOR8 DETECT ALARM OUTPUT	S	2	Level4	Level4
System Setting - RJ					
4800	RJ STAGE	S	2	Level4	Level4
4801	RJ(HI-HI)	S	2	Level4	Level4
4802	RJ(HI-LIMIT)	S	2	Level4	Level4
4803	RJ(PASS)	S	2	Level4	Level4
4804	RJ(LO-LIMIT)	S	2	Level4	Level4
4805	RJ(LO-LO)	S	2	Level4	Level4
4806	RJ(METAL)	S	2	Level4	Level4
4807	RJ(ERROR)	S	2	Level4	Level4
4808	RJ(EXT.NG)	S	2	Level4	Level4
4809	RJ(PASS2)	S	2	Level4	Level4
4810	RJ(X-RAY)	S	2	Level4	Level4
4811	RJ(METAL+ERROR)	S	2	Level4	Level4
4812	RJ (AUTO-CALC INELIGIBLE)	S	2	Level4	Level4
4813	RJ BIN FULL	S	2	Level4	Level4
4814	ABNORMAL RJ TIMER	S	2	Level4	Level4
4815	OBSERV NO WGH	S	2	Level4	Level4
4816	RJ DEVICE.	S	2	Level4	Level4
4817	RJ MOTION STAGE1	S	2	Level4	Level4
4818	RJ MOTION STAGE2	S	2	Level4	Level4
4819	RJ MOTION STAGE3	S	2	Level4	Level4
4820	RJ MOTION STAGE4	S	2	Level4	Level4
4821	RJ MOTION STAGE5	S	2	Level4	Level4
4822	RJ MOTION STAGE6	S	2	Level4	Level4
4823	DIVERter DELAY	S	2	Level4	Level4
4824	PRE RJ-MOTION NUM	S	2	Level4	Level4
4825	PRE RJ-MOTION DLY TIM	S	2	Level4	Level4
4826	PRE RJ-MOTION TIM	S	2	Level4	Level4
System Setting - Option					
4900	WEIGH PH	S	2	Level4	Level4
4901	RJ PH	S	2	Level4	Level4
4902	AVERAGE SYSTEM(ON/OFF)	S	2	Level4	Level4
4903	LIMITATION OF VALUE	S	2	Level4	Level4
4904	ZERO TRACKING	S	2	Level4	Level4

ADDRESS	Contents	Type	Byte	Ref. Auth	Edi. Auth
4905	AZERO DIFF AMOUNT (ZF1 FILTER)	S	2	Level4	Level4
4906	AZERO DIFF AMOUNT (ZF2 FILTER)	S	2	Level4	Level4
4907	AZERO DIFF AMOUNT (ZF3 FILTER)	S	2	Level4	Level4
4908	AZERO DIFF AMOUNT (ZF4 FILTER)	S	2	Level4	Level4
4909	AVERAGE ALARM(ON/OFF)	S	2	Level4	Level4
4910	MODIFY TARGET(ON/OFF)	S	2	Level4	Level4
4911	CONTINUANCE NG (ON/OFF)	S	2	Level4	Level4
4912	CONVEYOR AUTO STOP	S	2	Level4	Level4
4913	CONDITION OF AUTO-STOP	L	4	Level4	Level4
4915	NUM OF WEIGH RANK	S	2	Level4	Level4
4916	NONE	S	2	Level4	Level4
4917	D-CUT TIMER SETTING	S	2	Level4	Level4
4918	NONE	S	2	Level4	Level4
4919	NONE	S	2	Level4	Level4
4920	AUTO-CALC TIMING	S	2	Level4	Level4
4921	METAL DETECTOR	S	2	Level4	Level4
4922	LOGIC OF MD	S	2	Level4	Level4
4923	EXT.NG	S	2	Level4	Level4
4924	EXT.NG2 PRIORITY	S	2	Level4	Level4
4925	X-RAY	S	2	Level4	Level4
4926	NONE	S	2	Level4	Level4
4927	INSPECTOR PRIORITY 1	S	2	Level4	Level4
4928	INSPECTOR PRIORITY 2	S	2	Level4	Level4
4929	INSPECTOR PRIORITY 3	S	2	Level4	Level4
4930	INSPECTOR PRIORITY 4	S	2	Level4	Level4
4931	BCD OUTPUT	S	2	Level4	Level4
4932	PRG NO. OUTPUT SET	S	2	Level4	Level4
4933	NONE	S	2	Level4	Level4
4934	NONE	S	2	Level4	Level4
4935	A/D FILTER DELAY	S	2	Level4	Level4
4936	NONE	S	2	Level4	Level4
4937	NONE	S	2	Level4	Level4
4938	WIDTH OF WEIGH SIGNAL	S	2	Level4	Level4
4939	STOP BTN IN ONLINE	S	2	Level4	Level4
4940	OVER OK	S	2	Level4	Level4
4941	NONE	S	2	Level4	Level4
4942	NONE	S	2	Level4	Level4
4943	USE MOTOR PARAMETER	S	2	Level4	Level4
4944	NONE	S	2	Level4	Level4
4945	CP/CPK CALC	S	2	Level4	Level4
4946	NONE	S	2	Level4	Level4
⋮				Level4	Level4
4949	NONE	S	2	Level4	Level4
4950	KTM RATIO	S	2	Level4	Level4
4951	MONITOR SLEEP TIME	S	2	Level4	Level4
4952	NONE	L	4	Level4	Level4
4954	MOTOR SLEEP TIMER	S	2	Level4	Level4
4955	BIN COVER ALARM	S	2	Level4	Level4

ADDRESS	Contents	Type	Byte	Ref. Auth	Edi. Auth
4956	NONE	S	2	Level4	Level4
4957	DW FEEDBACK(ON/OFF)	S	2	Level4	Level4
4958	NONE	S	2	Level4	Level4
4959	NONE	S	2	Level4	Level4
4960	NONE	L	4	Level4	Level4
4962	BIN LOCK	S	2	Level4	Level4
4963	MTL BIN LOCK	S	2	Level4	Level4
4964	BIN ABSENT	S	2	Level4	Level4
4965	MTL BIN ABSENT	S	2	Level4	Level4
4966	UNLOCK BIN TIME	S	2	Level4	Level4
4967	UNLOCK MTL BIN TIME	S	2	Level4	Level4
4968	BIN ABSENT FAULT	S	2	Level4	Level4
4969	MTL BIN ABSENT FAULT	S	2	Level4	Level4
4970	UNLOCK BIN PASSWORD	L	4	Level4	Level4
4972	UNLOCK MTL BIN PASSWORD	L	4	Level4	Level4
4974	DYNAMIC RANGE (LO)	S	2	Level4	Level4
4975	DYNAMIC RANGE (HI)	S	2	Level4	Level4
4976	MD TYPE	S	2	Level4	Level4
4977	OUTFEED BLOCKED SIGNAL	S	2	Level4	Level4
4978	FORCED RJ LENGTH	S	2	Level4	Level4
4979	MD READY	S	2	Level4	Level4
4980	PH LOGIC INVERSION	S	2	Level4	Level4
4981	DETECT INVAL. WT. (ON/OFF)	S	2	Level4	Level4
4982	NONE	L	4	Level4	Level4
4984	NONE	S	2	Level4	Level4
4985	MTR ALT DLY (COVER GRD OPEN / EMGSTOP)	S	2	Level4	Level4
4986	MTR ALT DLY (STATIC)	S	2	Level4	Level4
4987	INDIVIDUAL WEIGHT TWO LINE PRINT	S	2	Level4	Level4
4988	CONVEYOR ON CONDITION	S	2	Level4	Level4
4989	DISP AUTO ZERO	S	2	Level4	Level4
4990	LINEAR SPAN	S	2	Level4	Level4
4991	WEIGHT FLUCTUATION PREVENTION	S	2	Level4	Level4
4992	BATCH CODE INPUT	S	2	Level4	Level4
4993	NONE	S	2	Level4	Level4
4994	LENGTH LIM	S	2	Level4	Level4
4995	NONE	S	2	Level4	Level4
:				Level4	Level4
4997	NONE	S	2	Level4	Level4
System Setting - Special Option					
5100	NONE	S	2	Level4	Level4
:				Level4	Level4
5102	NONE	S	2	Level4	Level4
5103	MID MODE	S	2	Level4	Level4
5104	"e" MARK	S	2	Level4	Level4
5105	CHG MAX PROGRAM NUM	S	2	Level4	Level4
System Setting - Print					
5700	PRINTER MODEL	S	2	Level4	Level4
5701	CLEAR TOTAL WHEN CHG PRG SETTING	S	2	Level4	Level4

ADDRESS	Contents	Type	Byte	Ref. Auth	Edi. Auth
5702	CLEAR TOTAL WHEN CHG PRT SETTING	S	2	Level4	Level4
5703	MACHINE CODE1	L	4	Level2~	Level4
5705	MACHINE CODE2	L	4	Level2~	Level4
5707	MACHINE CODE3	L	4	Level2~	Level4
5709	MACHINE NAME1	C*7	14	Level2~	Level4
5716	MACHINE NAME2	C*7	14	Level2~	Level4
5723	MACHINE NAME3	C*7	14	Level2~	Level4
5730	LOT NUM PRT	S	2	Level2~	Level2~
5731	LOT NUM HEADER	C*7	14	Level2~	Level4
5738	OUTPUT DESTINATION	S	2	Level2~	Level2~
5739	HISTOGRAM STEP	L	4	Level2~	Level2~
5741	HISTOGRAM PRINT	S	2	Level2~	Level2~
5742	OUTPUT TTL INTERVAL (hour)	S	2	Level2~	Level2~
5743	OUTPUT TTL INTERVAL (min.)	S	2	Level2~	Level2~
5744	PRT TTL END OF LOG	S	2	Level2~	Level2~
5745	SAMPLE LOG TIME (hour)	S	2	Level2~	Level2~
5746	SAMPLE LOG TIME (min.)	S	2	Level2~	Level2~
5747	SAMPLE LOG PCS	L	4	Level2~	Level2~
5749	PRT TTL END OF SHIFT FLAG1	S	2	Level2~	Level2~
5750	PRT TTL END OF SHIFT FLAG2	S	2	Level2~	Level2~
5751	PRT TTL END OF SHIFT FLAG3	S	2	Level2~	Level2~
5752	PRT TTL END OF SHIFT1 (hour)	S	2	Level2~	Level2~
5753	PRT TTL END OF SHIFT1 (min.)	S	2	Level2~	Level2~
5754	PRT TTL END OF SHIFT2 (hour)	S	2	Level2~	Level2~
5755	PRT TTL END OF SHIFT2 (min.)	S	2	Level2~	Level2~
5756	PRT TTL END OF SHIFT3 (hour)	S	2	Level2~	Level2~
5757	PRT TTL END OF SHIFT3 (min.)	S	2	Level2~	Level2~
5758	AVG GRAPH SAMPLE NUM	L	4	Level2~	Level2~
5760	OUTPUT DEVICE	S	2	Level2~	Level2~
5761	LOT NUMBER	C*7	14	Level2~	Level2~
System Setting - Input Signal					
5900	INPUT SIGNAL1	S	2	Level4	Level4
5901	INPUT SIGNAL2	S	2	Level4	Level4
5902	INPUT SIGNAL3	S	2	Level4	Level4
5903	INPUT SIGNAL4	S	2	Level4	Level4
5904	INPUT SIGNAL5	S	2	Level4	Level4
5905	INPUT SIGNAL6	S	2	Level4	Level4
5906	INPUT SIGNAL7	S	2	Level4	Level4
5907	INPUT SIGNAL8	S	2	Level4	Level4
5908	INPUT SIGNAL9	S	2	Level4	Level4
5909	INPUT SIGNAL10	S	2	Level4	Level4
5910	INPUT SIGNAL11	S	2	Level4	Level4
5911	INPUT SIGNAL12	S	2	Level4	Level4
5912	INPUT SIGNAL13	S	2	Level4	Level4
5913	INPUT SIGNAL14	S	2	Level4	Level4
5914	INPUT SIGNAL15	S	2	Level4	Level4
5915	INPUT SIGNAL16	S	2	Level4	Level4
5916	INPUT LOGIC	L	4	Level4	Level4

ADDRESS	Contents	Type	Byte	Ref. Auth	Edi. Auth
System Setting Relay Output					
5918	RELAY SIGNAL1 OUTPUT1	S	2	Level4	Level4
5919	RELAY SIGNAL1 OUTPUT2	S	2	Level4	Level4
5920	RELAY SIGNAL1 OUTPUT3	S	2	Level4	Level4
5921	RELAY SIGNAL1 LOGIC1	S	2	Level4	Level4
5922	RELAY SIGNAL1 LOGIC2	S	2	Level4	Level4
5923	RELAY SIGNAL1 DELAY TIMER	L	4	Level4	Level4
5925	RELAY SIGNAL1 ACTION TIMER	L	4	Level4	Level4
5927	RELAY SIGNAL2 OUTPUT1	S	2	Level4	Level4
5928	RELAY SIGNAL2 OUTPUT2	S	2	Level4	Level4
5929	RELAY SIGNAL2 OUTPUT3	S	2	Level4	Level4
5930	RELAY SIGNAL2 LOGIC1	S	2	Level4	Level4
5931	RELAY SIGNAL2 LOGIC2	S	2	Level4	Level4
5932	RELAY SIGNAL2 DELAY TIMER	L	4	Level4	Level4
5934	RELAY SIGNAL2 ACTION TIMER	L	4	Level4	Level4
5936	RELAY SIGNAL3 OUTPUT1	S	2	Level4	Level4
5937	RELAY SIGNAL3 OUTPUT2	S	2	Level4	Level4
5938	RELAY SIGNAL3 OUTPUT3	S	2	Level4	Level4
5939	RELAY SIGNAL3 LOGIC1	S	2	Level4	Level4
5940	RELAY SIGNAL3 LOGIC2	S	2	Level4	Level4
5941	RELAY SIGNAL3 DELAY TIMER	L	4	Level4	Level4
5943	RELAY SIGNAL3 ACTION TIMER	L	4	Level4	Level4
5945	RELAY SIGNAL4 OUTPUT1	S	2	Level4	Level4
5946	RELAY SIGNAL4 OUTPUT2	S	2	Level4	Level4
5947	RELAY SIGNAL4 OUTPUT3	S	2	Level4	Level4
5948	RELAY SIGNAL4 LOGIC1	S	2	Level4	Level4
5949	RELAY SIGNAL4 LOGIC2	S	2	Level4	Level4
5950	RELAY SIGNAL4 DELAY TIMER	L	4	Level4	Level4
5952	RELAY SIGNAL4 ACTION TIMER	L	4	Level4	Level4
5954	RELAY SIGNAL5 OUTPUT1	S	2	Level4	Level4
5955	RELAY SIGNAL5 OUTPUT2	S	2	Level4	Level4
5956	RELAY SIGNAL5 OUTPUT3	S	2	Level4	Level4
5957	RELAY SIGNAL5 LOGIC1	S	2	Level4	Level4
5958	RELAY SIGNAL5 LOGIC2	S	2	Level4	Level4
5959	RELAY SIGNAL5 DELAY TIMER	L	4	Level4	Level4
5961	RELAY SIGNAL5 ACTION TIMER	L	4	Level4	Level4
5963	RELAY SIGNAL6 OUTPUT1	S	2	Level4	Level4
5964	RELAY SIGNAL6 OUTPUT2	S	2	Level4	Level4
5965	RELAY SIGNAL6 OUTPUT3	S	2	Level4	Level4
5966	RELAY SIGNAL6 LOGIC1	S	2	Level4	Level4
5967	RELAY SIGNAL6 LOGIC2	S	2	Level4	Level4
5968	RELAY SIGNAL6 DELAY TIMER	L	4	Level4	Level4
5970	RELAY SIGNAL6 ACTION TIMER	L	4	Level4	Level4
5972	RELAY SIGNAL7 OUTPUT1	S	2	Level4	Level4
5973	RELAY SIGNAL7 OUTPUT2	S	2	Level4	Level4
5974	RELAY SIGNAL7 OUTPUT3	S	2	Level4	Level4
5975	RELAY SIGNAL7 LOGIC1	S	2	Level4	Level4
5976	RELAY SIGNAL7 LOGIC2	S	2	Level4	Level4

ADDRESS	Contents	Type	Byte	Ref. Auth	Edi. Auth
5977	RELAY SIGNAL7 DELAY TIMER	L	4	Level4	Level4
5979	RELAY SIGNAL7 ACTION TIMER	L	4	Level4	Level4
5981	RELAY SIGNAL8 OUTPUT1	S	2	Level4	Level4
5982	RELAY SIGNAL8 OUTPUT2	S	2	Level4	Level4
5983	RELAY SIGNAL8 OUTPUT3	S	2	Level4	Level4
5984	RELAY SIGNAL8 LOGIC1	S	2	Level4	Level4
5985	RELAY SIGNAL8 LOGIC2	S	2	Level4	Level4
5986	RELAY SIGNAL8 DELAY TIMER	L	4	Level4	Level4
5988	RELAY SIGNAL8 ACTION TIMER	L	4	Level4	Level4
System Setting - IC Output					
5990	OUTPUT SIGNAL1 OUTPUT1	S	2	Level4	Level4
5991	OUTPUT SIGNAL1 OUTPUT2	S	2	Level4	Level4
5992	OUTPUT SIGNAL1 OUTPUT3	S	2	Level4	Level4
5993	OUTPUT SIGNAL1 LOGIC1	S	2	Level4	Level4
5994	OUTPUT SIGNAL1 LOGIC2	S	2	Level4	Level4
5997	OUTPUT SIGNAL1 DELAY TIMER	L	4	Level4	Level4
5995	OUTPUT SIGNAL1 ACTION TIMER	L	4	Level4	Level4
5999	OUTPUT SIGNAL2 OUTPUT1	S	2	Level4	Level4
6000	OUTPUT SIGNAL2 OUTPUT2	S	2	Level4	Level4
6001	OUTPUT SIGNAL2 OUTPUT3	S	2	Level4	Level4
6002	OUTPUT SIGNAL2 LOGIC1	S	2	Level4	Level4
6003	OUTPUT SIGNAL2 LOGIC2	S	2	Level4	Level4
6004	OUTPUT SIGNAL2 DELAY TIMER	L	4	Level4	Level4
6006	OUTPUT SIGNAL2 ACTION TIMER	L	4	Level4	Level4
6008	OUTPUT SIGNAL3 OUTPUT1	S	2	Level4	Level4
6009	OUTPUT SIGNAL3 OUTPUT2	S	2	Level4	Level4
6010	OUTPUT SIGNAL3 OUTPUT3	S	2	Level4	Level4
6011	OUTPUT SIGNAL3 LOGIC1	S	2	Level4	Level4
6012	OUTPUT SIGNAL3 LOGIC2	S	2	Level4	Level4
6013	OUTPUT SIGNAL3 DELAY TIMER	L	4	Level4	Level4
6015	OUTPUT SIGNAL3 ACTION TIMER	L	4	Level4	Level4
6017	OUTPUT SIGNAL4 OUTPUT1	S	2	Level4	Level4
6018	OUTPUT SIGNAL4 OUTPUT2	S	2	Level4	Level4
6019	OUTPUT SIGNAL4 OUTPUT3	S	2	Level4	Level4
6020	OUTPUT SIGNAL4 LOGIC1	S	2	Level4	Level4
6021	OUTPUT SIGNAL4 LOGIC2	S	2	Level4	Level4
6022	OUTPUT SIGNAL4 DELAY TIMER	L	4	Level4	Level4
6024	OUTPUT SIGNAL4 ACTION TIMER	L	4	Level4	Level4
6026	OUTPUT SIGNAL5 OUTPUT1	S	2	Level4	Level4
6027	OUTPUT SIGNAL5 OUTPUT2	S	2	Level4	Level4
6028	OUTPUT SIGNAL5 OUTPUT3	S	2	Level4	Level4
6029	OUTPUT SIGNAL5 LOGIC1	S	2	Level4	Level4
6030	OUTPUT SIGNAL5 LOGIC2	S	2	Level4	Level4
6031	OUTPUT SIGNAL5 DELAY TIMER	L	4	Level4	Level4
6033	OUTPUT SIGNAL5 ACTION TIMER	L	4	Level4	Level4
6035	OUTPUT SIGNAL6 OUTPUT1	S	2	Level4	Level4
6036	OUTPUT SIGNAL6 OUTPUT2	S	2	Level4	Level4
6037	OUTPUT SIGNAL6 OUTPUT3	S	2	Level4	Level4

ADDRESS	Contents	Type	Byte	Ref. Auth	Edi. Auth
6038	OUTPUT SIGNAL6 LOGIC1	S	2	Level4	Level4
6039	OUTPUT SIGNAL6 LOGIC2	S	2	Level4	Level4
6040	OUTPUT SIGNAL6 DELAY TIMER	L	4	Level4	Level4
6042	OUTPUT SIGNAL6 ACTION TIMER	L	4	Level4	Level4
6044	OUTPUT SIGNAL7 OUTPUT1	S	2	Level4	Level4
6045	OUTPUT SIGNAL7 OUTPUT2	S	2	Level4	Level4
6046	OUTPUT SIGNAL7 OUTPUT3	S	2	Level4	Level4
6047	OUTPUT SIGNAL7 LOGIC1	S	2	Level4	Level4
6048	OUTPUT SIGNAL7 LOGIC2	S	2	Level4	Level4
6049	OUTPUT SIGNAL7 DELAY TIMER	L	4	Level4	Level4
6051	OUTPUT SIGNAL7 ACTION TIMER	L	4	Level4	Level4
6053	OUTPUT SIGNAL8 OUTPUT1	S	2	Level4	Level4
6054	OUTPUT SIGNAL8 OUTPUT2	S	2	Level4	Level4
6055	OUTPUT SIGNAL8 OUTPUT3	S	2	Level4	Level4
6056	OUTPUT SIGNAL8 LOGIC1	S	2	Level4	Level4
6057	OUTPUT SIGNAL8 LOGIC2	S	2	Level4	Level4
6058	OUTPUT SIGNAL8 DELAY TIMER	L	4	Level4	Level4
6060	OUTPUT SIGNAL8 ACTION TIMER	L	4	Level4	Level4
System Setting - Error Allocation					
6600	CONDITION OF COV	L	4	Level4	Level4
6602	USER ERROR TEXT1	S*14	28	Level4	Level4
6616	NONE	S	2	Level4	Level4
6617	NONE	S	2	Level4	Level4
6618	USER ERROR TEXT2	S*14	28	Level4	Level4
6632	NONE	S	2	Level4	Level4
6633	NONE	S	2	Level4	Level4
6634	USER ERROR TEXT3	S*14	28	Level4	Level4
6648	NONE	S	2	Level4	Level4
6649	NONE	S	2	Level4	Level4
6650	USER ERROR TEXT4	S*14	28	Level4	Level4
6664	NONE	S	2	Level4	Level4
6665	NONE	S	2	Level4	Level4
6666	CONDITION OF PWD ERROR	L	4	Level4	Level4
6668	CONDITION OF CORRECT PWD ERROR	L	4	Level4	Level4
6670	LIM OF CONTROLLED RE-START PASSWORD	S	2	Level4	Level4
System Setting - Barcode					
6800	BARCODE DETECTION DELAY TIMER	S	2	Level4	Level4
6801	BARCODE DETECTION DO TIMER	S	2	Level4	Level4
6802	BARCODE JUDGE DELAY TIMER	S	2	Level4	Level4
6803	BARCODE READ START POSITION	S	2	Level4	Level4
6804	BARCODE READ NUM OF DIGITS	S	2	Level4	Level4
Program Setting - Basic					
8000	PROGRAM NAME	S*10	20	ALL	Level2~
8020	INFD. SPD	S	2	ALL	Level2~
8021	LENGTH	S	2	ALL	Level2~
8022	INFEED CONVEYOR SPEED	S	2	ALL	Level2~
8023	WEIGH CONVEYOR SPEED	S	2	ALL	Level2~
8024	RJ CONVEYOR SPEED	S	2	ALL	Level2~

ADDRESS	Contents	Type	Byte	Ref. Auth	Edi. Auth
8025	OPTION1 CONVEYOR SPEED	S	2	ALL	Level2~
8026	OPTION2 CONVEYOR SPEED	S	2	ALL	Level2~
8027	OPTION3 CONVEYOR SPEED	S	2	ALL	Level2~
8028	OPTION4 CONVEYOR SPEED	S	2	ALL	Level2~
8029	OPTION5 CONVEYOR SPEED	S	2	ALL	Level2~
8030	NONE	S	2	ALL	Level2~
8031	NONE	S-	2	ALL	Level2~
Program Setting - Weigh Rank					
8100	AVERAGE SYSTEM MODE	L	4	ALL	Level2~
8102	PASS	L	4	ALL	Level2~
8104	HI-HI	L	4	ALL	Level2~
8106	HI-LIMIT	L	4	ALL	Level2~
8108	LO-LIMIT	L	4	ALL	Level2~
8110	LO-LO	L	4	ALL	Level2~
8112	HI -LIMIT IN PASS	L	4	ALL	Level2~
8114	LO-LIMIT IN PASS	L	4	ALL	Level2~
8116	NOMINAL	L	4	ALL	Level2~
8118	SHIFT WEIGH	L	4	ALL	Level2~
8120	T1	L	4	ALL	DENIED
8122	T2	L	4	ALL	DENIED
Program Setting - RJ					
8200	RJ DELAY TIMER(STAGE1)	S	2	ALL	Level2~
8201	RJ TIMER (STAGE1)	S	2	ALL	Level2~
8202	RJ CONF. DELAY TIMER (STAGE1)	S	2	ALL	Level2~
8203	RJ CONF. TIMER(STAGE1)	S	2	ALL	Level2~
8204	RJ DELAY TIMER (STAGE2)	S	2	ALL	Level2~
8205	RJ TIMER (STAGE2)	S	2	ALL	Level2~
8206	RJ CONF. DELAY TIMER (STAGE2)	S	2	ALL	Level2~
8207	RJ CONF. TIMER (STAGE2)	S	2	ALL	Level2~
8208	RJ DELAY TIMER (STAGE3)	S	2	ALL	Level2~
8209	RJ TIMER (STAGE3)	S	2	ALL	Level2~
8210	RJ CONF. DELAY TIMER (STAGE3)	S	2	ALL	Level2~
8211	RJ CONF. TIMER (STAGE3)	S	2	ALL	Level2~
8212	RJ DELAY TIMER (STAGE4)	S	2	ALL	Level2~
8213	RJ TIMER (STAGE4)	S	2	ALL	Level2~
8214	RJ CONF. DELAY TIMER (STAGE4)	S	2	ALL	Level2~
8215	RJ CONF. TIMER (STAGE4)	S	2	ALL	Level2~
8216	RJ DELAY TIMER (STAGE5)	S	2	ALL	Level2~
8217	RJ TIMER (STAGE5)	S	2	ALL	Level2~
8218	RJ CONF. DELAY TIMER (STAGE5)	S	2	ALL	Level2~
8219	RJ CONF. TIMER (STAGE5)	S	2	ALL	Level2~
8220	RJ DELAY TIMER (STAGE6)	S	2	ALL	Level2~
8221	RJ TIMER (STAGE6)	S	2	ALL	Level2~
8222	RJ CONF. DELAY TIMER (STAGE6)	S	2	ALL	Level2~
8223	RJ CONF. TIMER (STAGE6)	S	2	ALL	Level2~

ADDRESS	Contents	Type	Byte	Ref. Auth	Edi. Auth
Program Setting – Filter					
8300	NONE	S	2	ALL	Level3~
8301	DYNAMIC FILTER1	S	2	ALL	Level3~
8302	DYNAMIC FILTER2	S	2	ALL	Level3~
8303	DYNAMIC FILTER3	S	2	ALL	Level3~
8304	DYNAMIC FILTER4	S	2	ALL	Level3~
8305	NONE	S	2	ALL	Level3~
8306	NONE	S	2	ALL	Level3~
8307	ZERO FILTER1	S	2	ALL	Level3~
8308	ZERO FILTER2	S	2	ALL	Level3~
8309	ZERO FILTER3	S	2	ALL	Level3~
8310	ZERO FILTER4	S	2	ALL	Level3~
8311	NONE	S	2	ALL	Level3~
8312	NONE	S	2	ALL	Level3~
8313	INFD. SPD SETTING	S	2	ALL	Level3~
8314	ATM READY TIME	S	2	ALL	Level3~
8315	AUTO-ZERO WATCH TIME	S	2	ALL	Level3~
Program Setting - Weigh Timing					
8400	AUTO ZERO TIMING(ATM)	S	2	ALL	Level2~
8401	WEIGH TIMING(KTM)	S	2	ALL	Level2~
8402	PH1 DOUBLE CUT TIMER	S	2	ALL	Level2~
8403	PH2 DOUBLE CUT TIMER	S	2	ALL	Level2~
8404	NONE	S	2	ALL	Level2~
8405	RJ BIN-FULL TIMER	S	2	ALL	Level2~
8406	NONE	S	2	ALL	Level2~
8407	RJ SHIFT TIMER	S	2	ALL	Level2~
8408	NONE	S	2	ALL	Level2~
8409	ABNORMAL RJ TIMER	S	2	ALL	Level2~
8410	OUTFEED BLOCKED SENSOR	L	4	ALL	Level2~
8412	NONE	L	4	ALL	Level2~
8414	AUTO ZERO TIMING (ATM2)	S	2	ALL	Level2~
Program Setting – RJ, PASS					
8500	CHG PASS RJ (WEIGH/NUM/OFF)	S	2	ALL	Level2~
8501	PULSE WIDTH	S	2	ALL	Level2~
8502	VALUE	L	4	ALL	Level2~
8504	CHG RJ PLS DLAY	S	2	ALL	Level2~
Program Setting - DYNAMIC COMP., PT					
8600	PT	L	4	ALL	Level2~
8602	NONE	L	4	ALL	Level2~
8604	DYNAMIC COMP.	L	4	ALL	Level2~
8606	NONE	L	4	ALL	Level2~
8608	NONE	L	4	ALL	Level2~
Program Setting - Ext.					
8700	LOGIC OF MD	S	2	ALL	Level2~
8701	RJ OVER	S	2	ALL	Level2~
Program Setting - Detect Invalid Weight					
8702	DETECT INVALID WEIGHT (ON/OFF)	S	2	ALL	Level2~

ADDRESS	Contents	Type	Byte	Ref. Auth	Edi. Auth
8703	DETECT INVALID WEIGHT VALUE	L	4	ALL	Level2~
8705	NONE	L	4	ALL	Level2~
8707	NONE	L	4	ALL	Level2~
Program Setting - Average Alarm					
8800	AVERAGE ALARM(ON/OFF)	S	2	ALL	Level2~
8801	AVERAGE ALARM PASS	L	4	ALL	Level2~
8803	AVERAGE ALARM HI-LIMIT	L	4	ALL	Level2~
8805	AVERAGE ALARM LO-LIMIT	L	4	ALL	Level2~
8807	AVERAGE ALARM SAMPLE NUM	L	4	ALL	Level2~
Program Setting – Feedback					
8900	FEED BACK (ON/OFF)	S	2	ALL	Level2~
8901	FEED BACK CALC NUM (SAMPLE NUM)	L	4	ALL	Level2~
8903	FEED BACK STANBY NUM	L	4	ALL	Level2~
8905	FEED BACK PASS	L	4	ALL	Level2~
8907	FEED BACK HI-LIMIT	L	4	ALL	Level2~
8909	FEED BACK LO-LIMIT	L	4	ALL	Level2~
8911	DEAD ZONE	L	4	ALL	Level2~
8913	FEED BACK EXCHG FACTOR (EXCHG RATE)	L	4	ALL	Level2~
8915	NONE	S	2	ALL	Level2~
8916	NONE	S	2	ALL	Level2~
Program Setting - Modify Target					
9000	MODIFY TARGET (ON/OFF)	S	2	ALL	Level2~
9001	NONE	L	4	ALL	Level2~
9003	LIMIT	L	4	ALL	Level2~
9005	SAMPLE NUM	L	4	ALL	Level2~
Program Setting - Continuance NG					
9100	CONTINUANCE NG (ON/OFF)	S	2	ALL	Level2~
9101	CONTINUANCE NG SAMPLE NUM	S	2	ALL	Level2~
Program Setting - DW Feedback					
9300	DW FEED BACK (ON/OFF)	S	2	ALL	Level2~
9301	DW FEED BACK SAMPLE NUM	L	4	ALL	Level2~
9303	DW FEED BACK STANBY NUM	L	4	ALL	Level2~
9305	DW FEED BACK PASS	L	4	ALL	Level2~
9307	DW FEED BACK HI-LIMIT	L	4	ALL	Level2~
9309	DW FEED BACK LO-LIMIT	L	4	ALL	Level2~
9311	DW FEED BACK EXCHG FACTOR	L	4	ALL	Level2~
9313	MIN LIMIT	L	4	ALL	Level2~
9315	MAX LIMIT	L	4	ALL	Level2~

4.4 Input Register

ADDRESS is originally from 40000 to 49999, but the highest level is omitted.

ADDRESS	Contents	Type	Byte	Ref. Auth
Display				
0100	WEIGH	L	4	ALL
0102	RESULT	L	4	ALL
0104	TOTAL CODE	L	4	ALL
0106	SUB-TOTAL CODE	L	4	ALL
0108	A/D COUNT	L	4	ALL
0110	PASS	L	4	ALL
0112	RLn	L	4	ALL
0114	EXT.NG FLAG	L	4	ALL
Error				
0200	NONE	L	4	ALL
0202	ERROR(GENERAL)	L	4	ALL
0204	ERROR (GENERAL)	L	4	ALL
0206	NONE	L	4	ALL
:				ALL
0216	NONE	L	4	ALL
0218	ERROR (discrete)	L	4	ALL
0220	ERROR (WT)	L	4	ALL
0222	NONE	L	4	ALL
:				ALL
0232	NONE	L	4	ALL
Total				
2100	MEASURE START TIME (year)	S	2	ALL
2101	MEASURE START TIME (mon.)	S	2	ALL
2102	MEASURE START TIME (day)	S	2	ALL
2103	MEASURE START TIME (hour)	S	2	ALL
2104	MEASURE START TIME (min.)	S	2	ALL
2105	MEASURE START TIME (sec.)	S	2	ALL
2106	MEASURE END TIME (year)	S	2	ALL
2107	MEASURE END TIME (mon.)	S	2	ALL
2108	MEASURE END TIME (day)	S	2	ALL
2109	MEASURE END TIME (hour)	S	2	ALL
2110	MEASURE END TIME (min.)	S	2	ALL
2111	MEASURE END TIME (sec.)	S	2	ALL
2112	HI-HI MEASURE NUM	L	4	ALL
2114	HI-HI MEASURE WEIGH	LL	8	ALL
2118	HI-LIMIT MEASURE NUM	L	4	ALL
2120	HI-LIMIT MEASURE WEIGH	LL	8	ALL
2124	PASS MEASURE NUM	L	4	ALL
2126	PASS MEASURE WEIGH	LL	8	ALL
2130	LO-LIMIT MEASURE NUM	L	4	ALL
2132	LO-LIMIT MEASURE WEIGH	LL	8	ALL

ADDRESS	Contents	Type	Byte	Ref. Auth
2136	LO-LO MEASURE NUM	L	4	ALL
2138	LO-LO MEASURE WEIGH	LL	8	ALL
2142	METAL MEASURE NUM	L	4	ALL
2144	NONE	LL	8	ALL
2148	ERROR MEASURE NUM	L	4	ALL
2150	NONE	LL	8	ALL
2154	EXT.NG ERROR MEASURE NUM	L	4	ALL
2156	NONE	LL	8	ALL
2160	X-RAY ERROR MEASURE NUM	L	4	ALL
2162	NONE	LL	8	ALL
2166	TOTAL MEASURE NUM	L	4	ALL
2168	TOTAL MEASURE WEIGH	LL	8	ALL
2172	PASS MEASURE NUM	L	4	ALL
2174	PASS MEASURE WEIGH	LL	8	ALL
2178	PASS RATE	L	4	ALL
2180	TOTAL AVG. WEIGH	L	4	ALL
2182	TOTAL S.D.	L	4	ALL
2184	TOTAL RANGE	L	4	ALL
2186	TOTAL MAX WEIGH	L	4	ALL
2188	TOTAL MIN WEIGH	L	4	ALL
2190	PASS AVG. WEIGH	L	4	ALL
2192	PASS S.D.	L	4	ALL
2194	PASS RANGE	L	4	ALL
2196	PASS MAX WEIGH	L	4	ALL
2198	PASS MIN WEIGH	L	4	ALL
2200	CP	L	4	ALL
2202	CPK	L	4	ALL
2204	T1-T2 %	L	4	ALL
2206	T1-T2 NUM	L	4	ALL
2208	LO-LIMIT~LIMIT-LO NUM	L	4	ALL
2210	LO-LIMIT~LIMIT-LO WEIGH	L	4	ALL
2212	LIMIT-LO~HI-LIMIT NUM	L	4	ALL
2214	LIMIT-LO~HI-LIMIT WEIGH	L	4	ALL
2216	HI-LIMIT~LIMIT-HI NUM	L	4	ALL
2218	HI-LIMIT~LIMIT-HI WEIGH	L	4	ALL
2220	HISTOGRAM STEP (RANK1)	L	4	ALL
2222	HISTOGRAM STEP (RANK2)	L	4	ALL
2224	HISTOGRAM STEP (RANK3)	L	4	ALL
2226	HISTOGRAM STEP (RANK4)	L	4	ALL
2228	HISTOGRAM STEP (RANK5)	L	4	ALL
2230	HISTOGRAM STEP (RANK6)	L	4	ALL
2232	HISTOGRAM STEP (RANK7)	L	4	ALL
2234	HISTOGRAM STEP (RANK8)	L	4	ALL
2236	HISTOGRAM STEP (RANK9)	L	4	ALL
2238	HISTOGRAM STEP (RANK10)	L	4	ALL

ADDRESS	Contents	Type	Byte	Ref. Auth
2240	HISTOGRAM STEP (RANK11)	L	4	ALL
2242	HISTOGRAM STEP (RANK12)	L	4	ALL
2244	HISTOGRAM STEP (RANK13)	L	4	ALL
2246	HISTOGRAM STEP (RANK14)	L	4	ALL
2248	HISTOGRAM STEP (RANK15)	L	4	ALL
2250	HISTOGRAM STEP (RANK16)	L	4	ALL
2252	HISTOGRAM STEP (RANK17)	L	4	ALL
2254	HISTOGRAM STEP (RANK18)	L	4	ALL
2256	HISTOGRAM STEP (RANK19)	L	4	ALL
2258	HISTOGRAM STEP (RANK20)	L	4	ALL
2260	HISTOGRAM STEP (RANK21)	L	4	ALL
Total (SAMPLE LOG)				
2300	SAMPLE LOG MEASURE START TIME (year)	S	2	ALL
2301	SAMPLE LOG MEASURE START TIME (mon.)	S	2	ALL
2302	SAMPLE LOG MEASURE START TIME (day)	S	2	ALL
2303	SAMPLE LOG MEASURE START TIME (hour)	S	2	ALL
2304	SAMPLE LOG MEASURE START TIME (min.)	S	2	ALL
2305	SAMPLE LOG MEASURE START TIME (sec.)	S	2	ALL
2306	SAMPLE LOG MEASURE END TIME (year)	S	2	ALL
2307	SAMPLE LOG MEASURE END TIME (mon.)	S	2	ALL
2308	SAMPLE LOG MEASURE END TIME (day)	S	2	ALL
2309	SAMPLE LOG MEASURE END TIME (hour)	S	2	ALL
2310	SAMPLE LOG MEASURE END TIME (min.)	S	2	ALL
2311	SAMPLE LOG MEASURE END TIME (sec.)	S	2	ALL
2312	SAMPLE LOG HI-HI MEASURE NUM	L	4	ALL
2314	SAMPLE LOG HI-HI MEASURE WEIGH	LL	8	ALL
2318	SAMPLE LOG HI-LIMIT MEASURE NUM	L	4	ALL
2320	SAMPLE LOG HI-LIMIT MEASURE WEIGH	LL	8	ALL
2324	SAMPLE LOG PASS MEASURE NUM	L	4	ALL
2326	SAMPLE LOG PASS MEASURE WEIGH	LL	8	ALL
2330	SAMPLE LOG LO-LIMIT MEASURE NUM	L	4	ALL
2332	SAMPLE LOG LO-LIMIT MEASURE WEIGH	LL	8	ALL
2336	SAMPLE LOG LO-LO MEASURE NUM	L	4	ALL
2338	SAMPLE LOG LO-LO MEASURE WEIGH	LL	8	ALL
2342	SAMPLE LOG META LMEASURE NUM	L	4	ALL
2344	NONE	LL	8	ALL
2348	SAMPLE LOG ERROR MEASURE NUM	L	4	ALL
2350	NONE	LL	8	ALL
2354	SAMPLE LOG EXT.ING ERROR MEASURE NUM	L	4	ALL
2356	NONE	LL	8	ALL
2360	SAMPLE LOG X-RAY ERROR MEASURE NUM	L	4	ALL
2362	NONE	LL	8	ALL
2366	SAMPLE LOG TOTAL MEASURE NUM	L	4	ALL
2368	SAMPLE LOG TOTAL MEASURE WEIGH	LL	8	ALL
2372	SAMPLE LOG PASS MEASURE NUM	L	4	ALL

ADDRESS	Contents	Type	Byte	Ref. Auth
2374	SAMPLE LOG PASS MEASURE WEIGH	LL	8	ALL
2378	SAMPLE LOG PASS RATE	L	4	ALL
2380	SAMPLE LOG TOTAL AVG. WEIGH	L	4	ALL
2382	SAMPLE LOG TOTAL S.D.	L	4	ALL
2384	SAMPLE LOG TOTAL RANGE	L	4	ALL
2386	SAMPLE LOG TOTAL MAX WEIGH	L	4	ALL
2388	SAMPLE LOG TOTAL MIN WEIGH	L	4	ALL
2390	SAMPLE LOG PASS AVG. WEIGH	L	4	ALL
2392	SAMPLE LOG PASS S.D.	L	4	ALL
2394	SAMPLE LOG PASS RANGE	L	4	ALL
2396	SAMPLE LOG PASS MAX WEIGH	L	4	ALL
2398	SAMPLE LOG PASS MIN WEIGH	L	4	ALL
2400	SAMPLE LOG CP	L	4	ALL
2402	SAMPLE LOG CPK	L	4	ALL
2404	SAMPLE LOG T1-T2 %	L	4	ALL
2406	SAMPLE LOG T1-T2 NUM	L	4	ALL
2408	SAMPLE LOG LO-LIMIT~LIMIT-LO NUM	L	4	ALL
2410	SAMPLE LOG LO-LIMIT~LIMIT-LO WEIGH	L	4	ALL
2412	SAMPLE LOG LIMIT-LO~HI-LIMIT NUM	L	4	ALL
2414	SAMPLE LOG LIMIT-LO~HI-LIMIT WEIGH	L	4	ALL
2416	SAMPLE LOG HI-LIMIT~LIMIT-HI NUM	L	4	ALL
2418	SAMPLE LOG HI-LIMIT~LIMIT-HI WEIGH	L	4	ALL
2420	SAMPLE LOG HISTOGRAM STEP (RANK1)	L	4	ALL
2422	SAMPLE LOG HISTOGRAM STEP (RANK2)	L	4	ALL
2424	SAMPLE LOG HISTOGRAM STEP (RANK3)	L	4	ALL
2426	SAMPLE LOG HISTOGRAM STEP (RANK4)	L	4	ALL
2428	SAMPLE LOG HISTOGRAM STEP (RANK5)	L	4	ALL
2430	SAMPLE LOG HISTOGRAM STEP (RANK6)	L	4	ALL
2432	SAMPLE LOG HISTOGRAM STEP (RANK7)	L	4	ALL
2434	SAMPLE LOG HISTOGRAM STEP (RANK8)	L	4	ALL
2436	SAMPLE LOG HISTOGRAM STEP (RANK9)	L	4	ALL
2438	SAMPLE LOG HISTOGRAM STEP (RANK10)	L	4	ALL
2440	SAMPLE LOG HISTOGRAM STEP (RANK11)	L	4	ALL
2442	SAMPLE LOG HISTOGRAM STEP (RANK12)	L	4	ALL
2444	SAMPLE LOG HISTOGRAM STEP (RANK13)	L	4	ALL
2446	SAMPLE LOG HISTOGRAM STEP (RANK14)	L	4	ALL
2448	SAMPLE LOG HISTOGRAM STEP (RANK15)	L	4	ALL
2450	SAMPLE LOG HISTOGRAM STEP (RANK16)	L	4	ALL
2452	SAMPLE LOG HISTOGRAM STEP (RANK17)	L	4	ALL
2454	SAMPLE LOG HISTOGRAM STEP (RANK18)	L	4	ALL
2456	SAMPLE LOG HISTOGRAM STEP (RANK19)	L	4	ALL
2458	SAMPLE LOG HISTOGRAM STEP (RANK20)	L	4	ALL
2460	SAMPLE LOG HISTOGRAM STEP (RANK21)	L	4	ALL
Total (AVERAGE SYSTEM)				
2500	AVERAGE SYSTEM MEASURE START TIME (year)	S	2	ALL

ADDRESS	Contents	Type	Byte	Ref. Auth
2501	AVERAGE SYSTEM MEASURE START TIME (mon.)	S	2	ALL
2502	AVERAGE SYSTEM MEASURE START TIME (day)	S	2	ALL
2503	AVERAGE SYSTEM MEASURE START TIME (hour)	S	2	ALL
2504	AVERAGE SYSTEM MEASURE START TIME (min.)	S	2	ALL
2505	AVERAGE SYSTEM MEASURE START TIME (sec.)	S	2	ALL
2506	AVERAGE SYSTEM MEASURE END TIME (year)	S	2	ALL
2507	AVERAGE SYSTEM MEASURE END TIME (mon.)	S	2	ALL
2508	AVERAGE SYSTEM MEASURE END TIME (day)	S	2	ALL
2509	AVERAGE SYSTEM MEASURE END TIME (hour)	S	2	ALL
2510	AVERAGE SYSTEM MEASURE END TIME (min.)	S	2	ALL
2511	AVERAGE SYSTEM MEASURE END TIME (sec.)	S	2	ALL
2512	AVERAGE SYSTEM HI-HI MEASURE NUM	L	4	ALL
2514	AVERAGE SYSTEM HI-HI MEASURE WEIGH	LL	8	ALL
2518	AVERAGE SYSTEM HI-LIMIT MEASURE NUM	L	4	ALL
2520	AVERAGE SYSTEM HI-LIMIT MEASURE WEIGH	LL	8	ALL
2524	AVERAGE SYSTEM PASS MEASURE NUM	L	4	ALL
2526	AVERAGE SYSTEM PASS MEASURE WEIGH	LL	8	ALL
2530	AVERAGE SYSTEM LO-LIMIT MEASURE NUM	L	4	ALL
2532	AVERAGE SYSTEM LO-LIMIT MEASURE WEIGH	LL	8	ALL
2536	AVERAGE SYSTEM LO-LO MEASURE NUM	L	4	ALL
2538	AVERAGE SYSTEM LO-LO MEASURE WEIGH	LL	8	ALL
2542	AVERAGE SYSTEM METAL MEASURE NUM	L	4	ALL
2544	NONE	LL	8	ALL
2548	AVERAGE SYSTEM ERROR MEASURE NUM	L	4	ALL
2550	NONE	LL	8	ALL
2554	AVERAGE SYSTEM EXT.NG. ERROR MEASURE NUM	L	4	ALL
2556	NONE	LL	8	ALL
2560	AVERAGE SYSTEM X-RAY ERROR MEASURE NUM	L	4	ALL
2562	NONE	LL	8	ALL
2566	AVERAGE SYSTEM TOTAL MEASURE NUM	L	4	ALL
2568	AVERAGE SYSTEM TOTAL MEASURE WEIGH	LL	8	ALL
2572	AVERAGE SYSTEM PASS MEASURE NUM	L	4	ALL
2574	AVERAGE SYSTEM PASS MEASURE WEIGH	LL	8	ALL
2578	AVERAGE SYSTEM PASS RATE	L	4	ALL
2580	AVERAGE SYSTEM TOTA LAVG. WEIGH	L	4	ALL
2582	AVERAGE SYSTEM TOTAL S.D.	L	4	ALL
2584	AVERAGE SYSTEM TOTAL RANGE	L	4	ALL
2586	AVERAGE SYSTEM TOTAL MAX WEIGH	L	4	ALL
2588	AVERAGE SYSTEM TOTAL MIN WEIGH	L	4	ALL
2590	AVERAGE SYSTEM PASS AVG. WEIGH	L	4	ALL
2592	AVERAGE SYSTEM PASS S.D.	L	4	ALL
2594	AVERAGE SYSTEM PASS	L	4	ALL
2596	AVERAGE SYSTEM PASS MAX WEIGH	L	4	ALL
2598	AVERAGE SYSTEM PASS MIN WEIGH	L	4	ALL
2600	AVERAGE SYSTEM CP	L	4	ALL

ADDRESS	Contents	Type	Byte	Ref. Auth
2602	AVERAGE SYSTEM CPK	L	4	ALL
2604	AVERAGE SYSTEM T1-T2 %	L	4	ALL
2606	AVERAGE SYSTEM T1-T2 NUM	L	4	ALL
2608	AVERAGE SYSTEM LO-LIMIT~LIMIT-LO NUM	L	4	ALL
2610	AVERAGE SYSTEM LO-LIMIT~LIMIT-LO WEIGH	L	4	ALL
2612	AVERAGE SYSTEM LIMIT-LO~HI-LIMIT NUM	L	4	ALL
2614	AVERAGE SYSTEM LIMIT-LO~HI-LIMIT WEIGH	L	4	ALL
2616	AVERAGE SYSTEM HI-LIMIT~LIMIT-HI NUM	L	4	ALL
2618	AVERAGE SYSTEM HI-LIMIT~LIMIT-HI WEIGH	L	4	ALL
2620	AVERAGE SYSTEM HISTOGRAM STEP (RANK1)	L	4	ALL
2622	AVERAGE SYSTEM HISTOGRAM STEP (RANK2)	L	4	ALL
2624	AVERAGE SYSTEM HISTOGRAM STEP (RANK3)	L	4	ALL
2626	AVERAGE SYSTEM HISTOGRAM STEP (RANK4)	L	4	ALL
2628	AVERAGE SYSTEM HISTOGRAM STEP (RANK5)	L	4	ALL
2630	AVERAGE SYSTEM HISTOGRAM STEP (RANK6)	L	4	ALL
2632	AVERAGE SYSTEM HISTOGRAM STEP (RANK7)	L	4	ALL
2634	AVERAGE SYSTEM HISTOGRAM STEP (RANK8)	L	4	ALL
2636	AVERAGE SYSTEM HISTOGRAM STEP (RANK9)	L	4	ALL
2638	AVERAGE SYSTEM HISTOGRAM STEP (RANK10)	L	4	ALL
2640	AVERAGE SYSTEM HISTOGRAM STEP (RANK11)	L	4	ALL
2642	AVERAGE SYSTEM HISTOGRAM STEP (RANK12)	L	4	ALL
2644	AVERAGE SYSTEM HISTOGRAM STEP (RANK13)	L	4	ALL
2646	AVERAGE SYSTEM HISTOGRAM STEP (RANK14)	L	4	ALL
2648	AVERAGE SYSTEM HISTOGRAM STEP (RANK15)	L	4	ALL
2650	AVERAGE SYSTEM HISTOGRAM STEP (RANK16)	L	4	ALL
2652	AVERAGE SYSTEM HISTOGRAM STEP (RANK17)	L	4	ALL
2654	AVERAGE SYSTEM HISTOGRAM STEP (RANK18)	L	4	ALL
2656	AVERAGE SYSTEM HISTOGRAM STEP (RANK19)	L	4	ALL
2658	AVERAGE SYSTEM HISTOGRAM STEP (RANK20)	L	4	ALL
2660	AVERAGE SYSTEM HISTOGRAM STEP (RANK21)	L	4	ALL
Total (EX.)				
2700	INDIVIDUAL PRT COUNTER	L	4	ALL
2702	RJ PASS NUM	L	4	ALL
2704	PASS NUM	L	4	ALL
Total (AVG.GRAPH)				
2800	AVG. GRAPH TOTAL WEIGH	L	4	ALL
2802	AVG. GRAPH MEASURE NUM	L	4	ALL
2804	AVG. GRAPH INDEX NUM	L	4	ALL
2806	AVG. OF NUM1	L	4	ALL
2808	AVG. OF NUM2	L	4	ALL
2810	AVG. OF NUM3	L	4	ALL
2812	AVG. OF NUM4	L	4	ALL
2814	AVG. OF NUM5	L	4	ALL
2816	AVG. OF NUM6	L	4	ALL
2818	AVG. OF NUM7	L	4	ALL

ADDRESS	Contents	Type	Byte	Ref. Auth
2820	AVG. OF NUM8	L	4	ALL
2822	AVG. OF NUM9	L	4	ALL
2824	AVG. OF NUM10	L	4	ALL
2826	AVG. OF NUM11	L	4	ALL
2828	AVG. OF NUM12	L	4	ALL
2830	AVG. OF NUM13	L	4	ALL
2832	AVG. OF NUM14	L	4	ALL
2834	AVG. OF NUM15	L	4	ALL
2836	AVG. OF NUM16	L	4	ALL
2838	AVG. OF NUM17	L	4	ALL
2840	AVG. OF NUM18	L	4	ALL
2842	AVG. OF NUM19	L	4	ALL
2844	AVG. OF NUM20	L	4	ALL
2846	AVG. OF NUM21	L	4	ALL
Total (INFD. SPD)				
2900	INFD. SPD	L	4	ALL
2902	INFD. SPD(PASS)	L	4	ALL
Individual Weight				
3000	INDIVIDUAL WEIGHT LEFT	S	2	ALL
3001	INDIVIDUAL WEIGHT MEASURE START TIME (year)	S	2	ALL
3002	INDIVIDUAL WEIGHT MEASURE START TIME (mon.)	S	2	ALL
3003	INDIVIDUAL WEIGHT MEASURE START TIME (day)	S	2	ALL
3004	INDIVIDUAL WEIGHT MEASURE START TIME (hour)	S	2	ALL
3005	INDIVIDUAL WEIGHT MEASURE START TIME (min.)	S	2	ALL
3006	INDIVIDUAL WEIGHT MEASURE START TIME (sec.)	S	2	ALL
3007	INDIVIDUAL WEIGHT GROUP CODE	S	2	ALL
3008	INDIVIDUAL WEIGHT PROGRAM NUM	S	2	ALL
3009	INDIVIDUAL WEIGHT SERIAL NUM	S	2	ALL
3011	INDIVIDUAL WEIGHT MEASURE IMPLEMENT TIME (year)	L	4	ALL
3012	INDIVIDUAL WEIGHT MEASURE IMPLEMENT TIME (mon.)	S	2	ALL
3013	INDIVIDUAL WEIGHT MEASURE IMPLEMENT TIME (day)	S	2	ALL
3014	INDIVIDUAL WEIGHT MEASURE IMPLEMENT TIME (hour)	S	2	ALL
3015	INDIVIDUAL WEIGHT MEASURE IMPLEMENT TIME (min.)	S	2	ALL
3016	INDIVIDUAL WEIGHT MEASURE IMPLEMENT TIME (sec.)	S	2	ALL
3017	RESULT WEIGH	S	2	ALL
3019	RESULT RANK	L	4	ALL
3020	RESULT UNIT	S	2	ALL
3021	RESULT DECIMAL POINT	S	2	ALL
Initial Calibration				
4200	NONE	S	2	Level4
4201	INITIAL COUNT	L	4	Level4
4203	NONE	LL	8	Level4
4207	GAIN/SPAN FACTOR (WEIGHT)	L	4	Level4
4209	GAIN/SPAN FACTOR (COUNT)	L	4	Level4
4211	SAMPLING WEIGHT1	L	4	Level4

ADDRESS	Contents	Type	Byte	Ref. Auth
4213	SAMPLING WEIGHT2	L	4	Level4
4215	SAMPLING WEIGHT3	L	4	Level4
4217	SAMPLING WEIGHT4	L	4	Level4
4219	SAMPLING COUNT1	L	4	Level4
4221	SAMPLING COUNT2	L	4	Level4
4223	SAMPLING COUNT3	L	4	Level4
4225	SAMPLING COUNT4	L	4	Level4
4227	NONE	L	4	Level4
4229	NONE	L	4	Level4
4231	QUADRATIC CORRECTION FACTOR1	D	8	Level4
4235	QUADRATIC CORRECTION FACTOR2	D	8	Level4
System Setting - RJ				
4800	RJ STAGE1	S	2	Level4
4801	RJ STAGE2	S	2	Level4
4802	RJ STAGE3	S	2	Level4
4803	RJ STAGE4	S	2	Level4
4804	RJ STAGE5	S	2	Level4
4805	RJ STAGE6	S	2	Level4
System Setting - Option				
4900	PROGRAM NUM INPUT	S	2	Level4
4901	FEED BACK	S	2	Level4
4902	RANDOM CHECKER	S	2	Level4
4903	RJ CONF.	S	2	Level4
4904	PROGRAM NUM INPUT	S	2	Level4
4905	PROGRAM NUM OUTPUT	S	2	Level4
4906	EXTERNAL DISPLAY	S	2	Level4
4907	Modbus/TCP	S	2	Level4
4908	BATCH CODE	C*2	4	Level4
System Setting - RS232C1(CN16)				
5200	RS232C1 PORT STATUS	S	2	Level4
5201	RS232C1 PROTOCOL	S	2	Level4
5202	RS232C1 BAUD RATE	S	2	Level4
5203	RS232C1 PARITY	S	2	Level4
5204	RS232C1 DATA LENGTH	S	2	Level4
5205	RS232C1 STOP BIT	S	2	Level4
5206	RS232C1 DATA KIND	S	2	Level4
5207	RS232C1 CR	S	2	Level4
System Setting - RS232C2(CN19)				
5300	RS232C2 PORT STATUS	S	2	Level4
5301	RS232C2 PROTOCOL	S	2	Level4
5302	RS232C2 BAUD RATE	S	2	Level4
5303	RS232C2 PARITY	S	2	Level4
5304	RS232C2 DATA LENGTH	S	2	Level4
5305	RS232C2 STOP BIT	S	2	Level4
5306	RS232C2 DATA KIND	S	2	Level4

ADDRESS	Contents	Type	Byte	Ref. Auth
5307	RS232C2 CR	S	2	Level4
System Setting - Ethernet				
5500	ETHERNET STATUS	S	2	Level4
5501	IP ADDRESS1(CW)	S	2	Level4
5502	IP ADDRESS2(CW)	S	2	Level4
5503	IP ADDRESS3(CW)	S	2	Level4
5504	IP ADDRESS4(CW)	S	2	Level4
5505	SUBNET MASK1	S	2	Level4
5506	SUBNET MASK2	S	2	Level4
5507	SUBNET MASK3	S	2	Level4
5508	SUBNET MASK4	S	2	Level4
5509	NAME ON NET	C*10	20	Level4
5519	PORT NUM (DW COM, CW SIDE)	S	2	Level4
5520	IP ADDRESS 1(DW COM, DW SIDE)	S	2	Level4
5521	IP ADDRESS 2(DW COM, DW SIDE)	S	2	Level4
5522	IP ADDRESS 3(DW COM, DW SIDE)	S	2	Level4
5523	IP ADDRESS 4(DW COM, DW SIDE)	S	2	Level4
5524	PORT NUM (DW COM, DW SIDE)	S	2	Level4
5525	PORT NUM (FORTRESS COM, CW SIDE)	S	2	Level4
5526	IP ADDRESS 1(FORTRESS COM, FORTRESS SIDE)	S	2	Level4
5527	IP ADDRESS 2(FORTRESS COM, FORTRESS SIDE)	S	2	Level4
5528	IP ADDRESS 3(FORTRESS COM, FORTRESS SIDE)	S	2	Level4
5529	IP ADDRESS 4(FORTRESS COM, FORTRESS SIDE)	S	2	Level4
5530	PORT NUM (FORTRESS COM, FORTRESS SIDE)	S	2	Level4
5531	EX DISPLAY PORT NUM	S	2	Level4
5532	Modbus/TCP PORT NUM	S	2	Level4
5533	GATEWAY1	S	2	Level4
5534	GATEWAY2	S	2	Level4
5535	GATEWAY3	S	2	Level4
5536	GATEWAY4	S	2	Level4
System Setting - Password				
6400	Level1 PASSWORD	C*5	10	Level4
6405	Level2~ PASSWORD	C*5	10	Level4
6410	Level3~ PASSWORD	C*5	10	Level4
6415	Level4 PASSWORD	C*5	10	Level4
6420	NONE	C*5	10	Level4
:				Level4
6430	NONE	C*5	10	Level4
System Setting - Error Allocation				
6600	ERROR ALLOCATION (Err1 ALARM)	L	4	Level4
6602	ERROR ALLOCATION (Err2 ALARM)	L	4	Level4
6604	NONE	L	4	Level4
:				Level4
6614	NONE	L	4	Level4
6616	ERROR ALLOCATION (Err9+10 ALARM)	LL	8	Level4

ADDRESS	Contents	Type	Byte	Ref. Auth
6620	ERROR ALLOCATION (WT ALARM)	L	4	Level4
6622	ERROR DIALOG FLAG (ALARM)	S	2	Level4
6623	NONE	S	2	Level4
6624	ERROR DIALOG CONTENTS (ALARM)	S	2	Level4
6625	ERROR ALLOCATION (Err1 FAULT)	L	4	Level4
6627	ERROR ALLOCATION (Err2FAULT)	L	4	Level4
6629	NONE	L	4	Level4
:				Level4
6639	NONE	L	4	Level4
6641	ERROR ALLOCATION (Err9+10FAULT)	LL	8	Level4
6645	ERROR ALLOCATION (WT FAULT)	L	4	Level4
6647	ERROR DIALOG FLAG (FAULT)	S	2	Level4
6648	NONE	S	2	Level4
6649	ERROR DIALOG CONTENTS (FAULT)	S	2	Level4
6650	ERROR ALLOCATION (Err1 ERROR1)	L	4	Level4
6652	ERROR ALLOCATION (Err2 ERROR1)	L	4	Level4
6654	NONE	L	4	Level4
:				Level4
6664	NONE	L	4	Level4
6666	ERROR ALLOCATION (Err9+10 ERROR1)	LL	8	Level4
6670	ERROR ALLOCATION (WT ERROR1)	L	4	Level4
6672	ERROR DIALOG FLAG (ERROR1)	S	2	Level4
6673	NONE	S	2	Level4
6674	ERROR DIALOG CONTENTS (ERROR1)	S	2	Level4
6675	ERROR ALLOCATION (Err1 ERROR2)	L	4	Level4
6677	ERROR ALLOCATION (Err2 ERROR2)	L	4	Level4
6679	NONE	L	4	Level4
:				Level4
6689	NONE	L	4	Level4
6691	ERROR ALLOCATION (Err9+10 ERROR2)	LL	8	Level4
6695	ERROR ALLOCATION (WT ERROR2)	L	4	Level4
6697	ERROR DIALOG FLAG (ERROR2)	S	2	Level4
6698	NONE	S	2	Level4
6699	ERROR DIALOG CONTENTS (ERROR2)	S	2	Level4
6700	ERROR ALLOCATION (Err1 ERROR3)	L	4	Level4
6702	ERROR ALLOCATION (Err2 ERROR3)	L	4	Level4
6704	NONE	L	4	Level4
:				Level4
6714	NONE	L	4	Level4
6716	ERROR ALLOCATION (Err9+10 ERROR3)	LL	8	Level4
6720	ERROR ALLOCATION (WT ERROR3)	L	4	Level4
6722	ERROR DIALOG FLAG (ERROR3)	S	2	Level4
6723	NONE	S	2	Level4
6724	ERROR DIALOG CONTENTS (ERROR3)	S	2	Level4
6725	ERROR ALLOCATION (Err1 ERROR4)	L	4	Level4

ADDRESS	Contents	Type	Byte	Ref. Auth
6727	ERROR ALLOCATION (Err2 ERROR4)	L	4	Level4
6729	NONE	L	4	Level4
6731	NONE	L	4	Level4
:				Level4
6739	NONE	L	4	Level4
6741	ERROR ALLOCATION (Err9+10 ERROR4)	LL	8	Level4
6745	ERROR ALLOCATION (WT ERROR4)	L	4	Level4
6747	ERROR DIALOG FLAG (ERROR4)	S	2	Level4
6748	NONE	S	2	Level4
6749	ERROR DIALOG CONTENTS (ERROR4)	S	2	Level4
Program setting - D.COMP, PT				
8600	D.COMP UPDATE DATE	L	4	ALL
8602	D.COMP UPDATE NUM	S	2	ALL